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Pengantar Kecerdasan  
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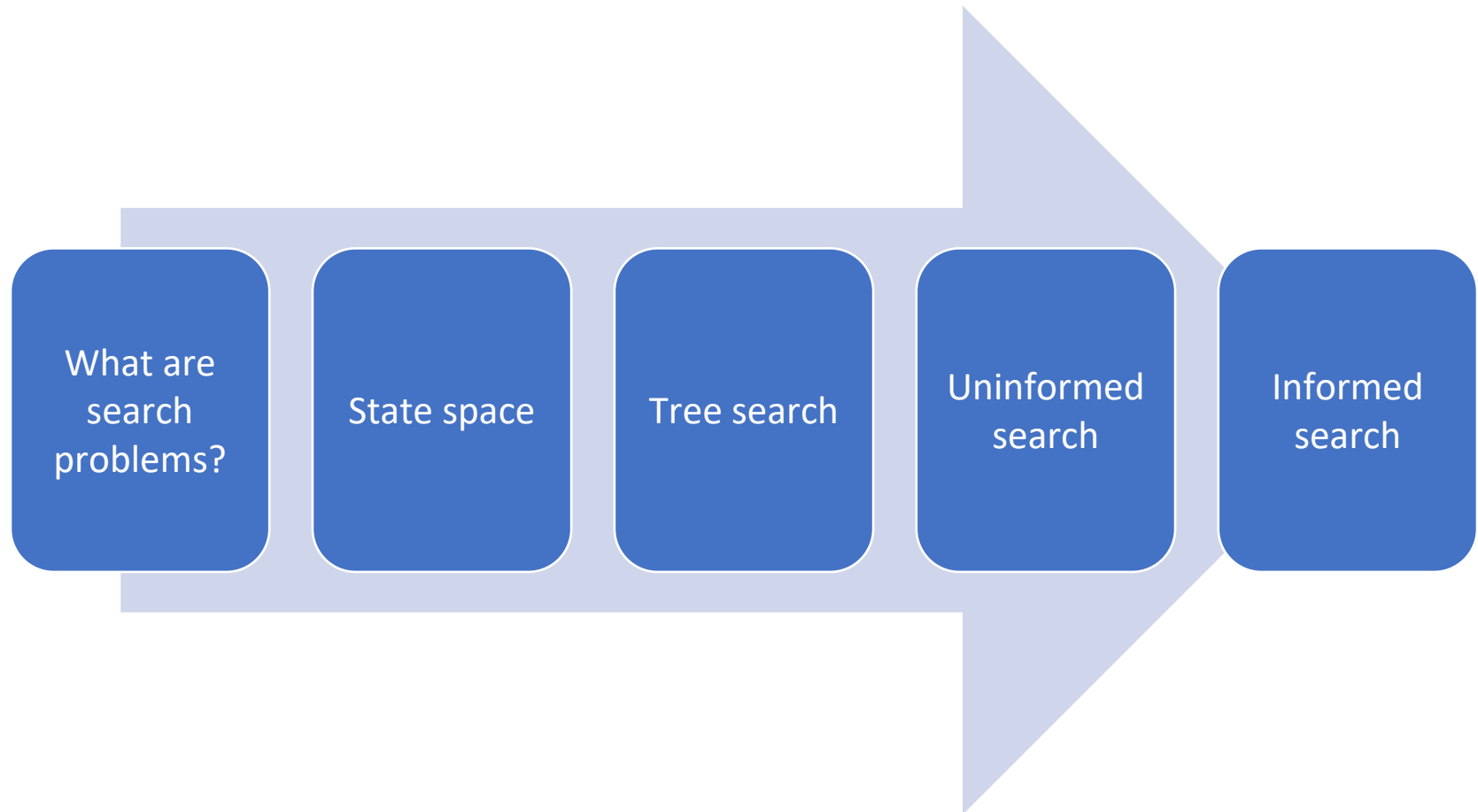
Solving problems  
by searching

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Bima Prihasto, Ph.D

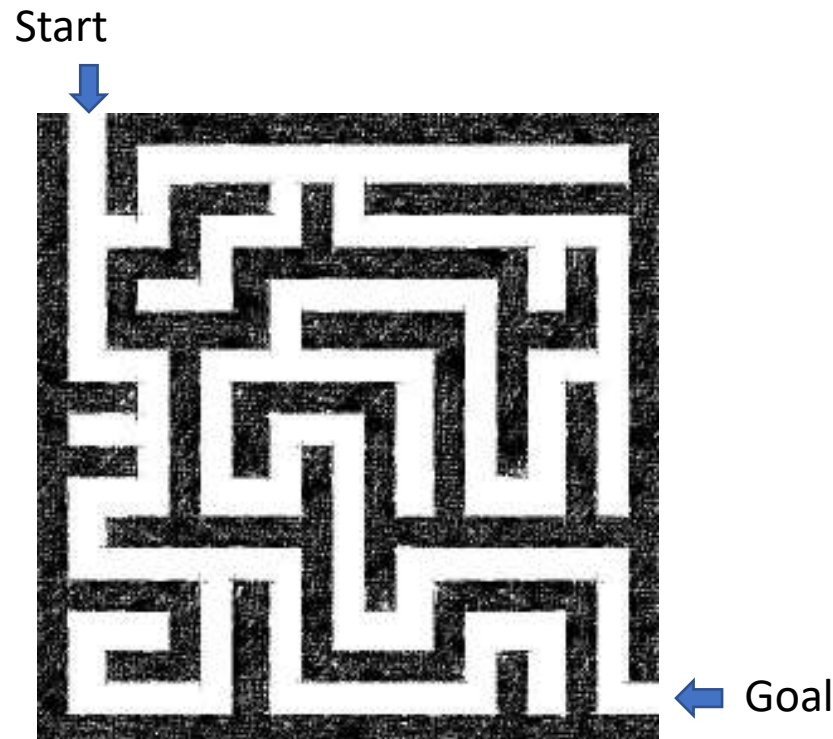
Prodi Informatika  
Institut Teknologi Kalimantan

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# What are search problems?

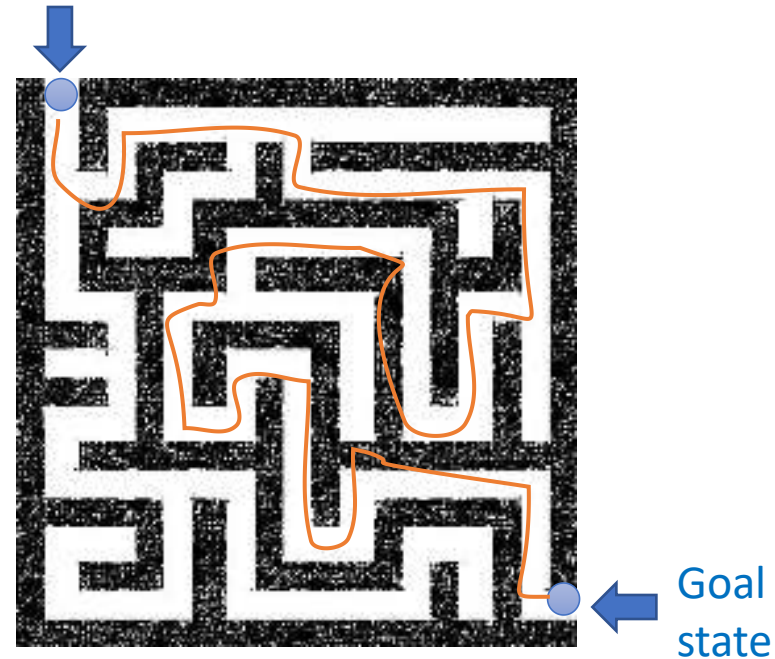
- We will consider the problem of designing **goal-based agents** in **known, fully observable, and deterministic** environments.
- Example:



# What are search problems?

- We will consider the problem of designing **goal-based agents** in , **known, fully observable, deterministic** environments.
- For now, we consider only a discrete environment using an **atomic state representation** (states are just labeled 1, 2, 3, ...).
- The **state space** is the set of all possible states of the environment and some states are marked as **goal states**.

Initial state

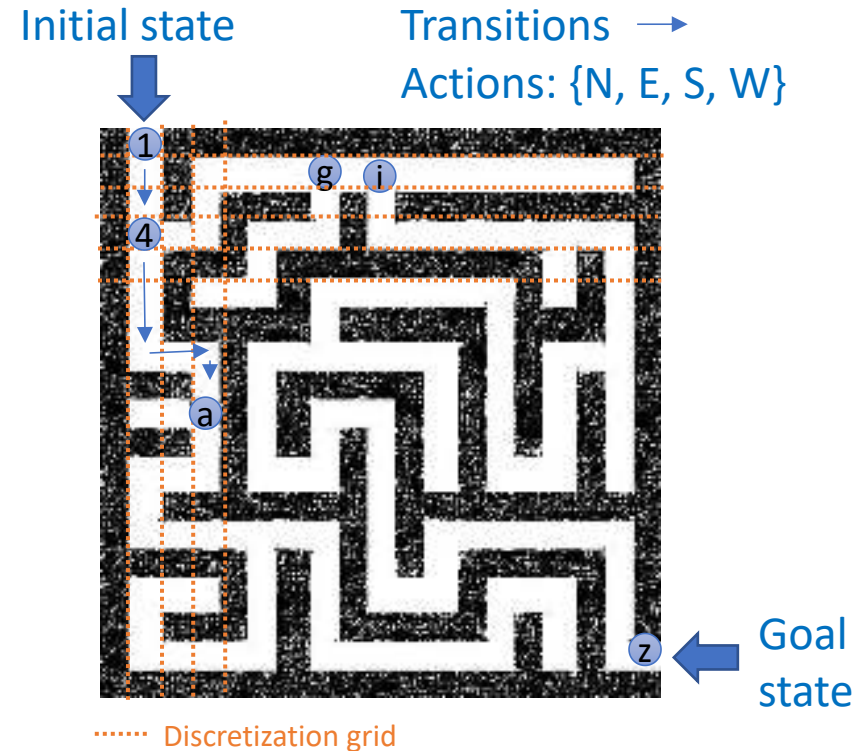


## Phases:

- 1) **Search**: the process of looking for the **sequence of actions** that reaches a goal state.
- 2) **Execution**: Once the agent begins executing the search solution in a deterministic, known environment, it can ignore its percepts (**open-loop system**).

# Search problem components

- **Initial state:** state description
- **Actions:** set of possible actions  $A$
- **Transition model:** a function that defines the new state resulting from performing an action in the current state  
 $f: S \times A \rightarrow S$  ( $S$  is the set of states)
- **Goal state:** state description
- **Path cost:** the sum of nonnegative *step costs*



## Notes:

- The **state space** is typically too large to be enumerated or it is continuous. Therefore, the problem is defined by initial state, actions and the transition model and not the set of all possible states.
- The **optimal solution** is the sequence of actions (or equivalently a sequence of states) that gives the lowest path cost for reaching the goal.

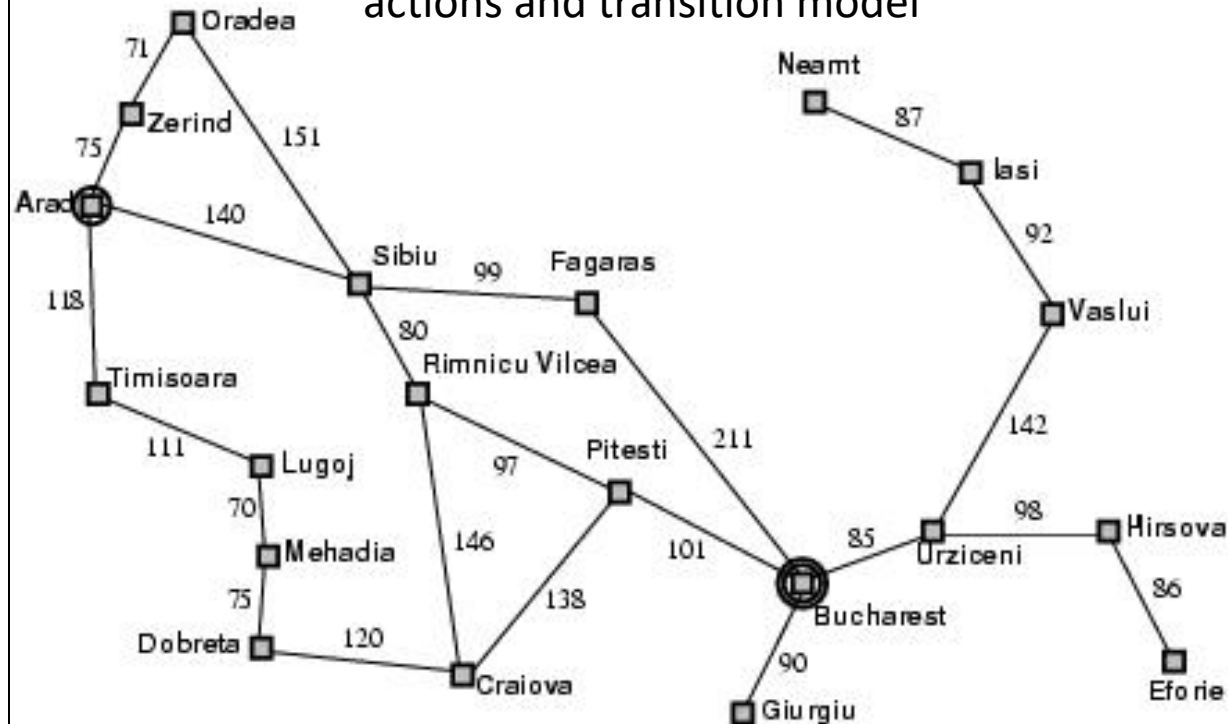
# Example: Romania Vacation

- On vacation in Romania; currently in Arad
- Flight leaves tomorrow from Bucharest



- **Initial state:** Arad
- **Actions:** Drive from one city to another.
- **Transition model and states:** If you go from city A to city B, you end up in city B.
- **Goal state:** Bucharest
- **Path cost:** Sum of edge costs.

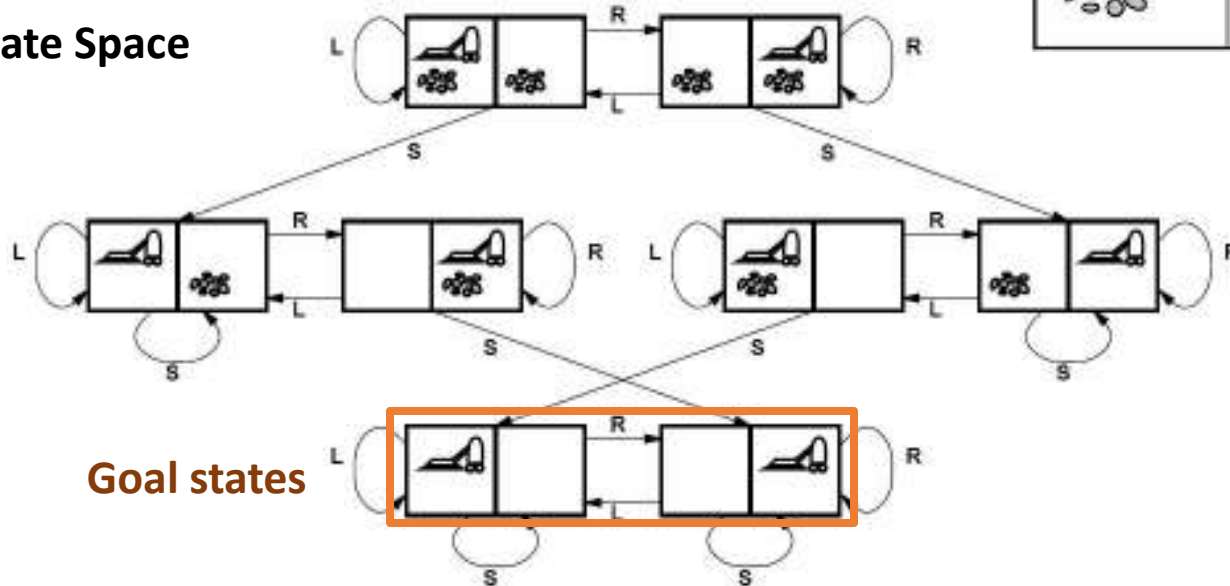
**State Space:** Defined by initial state, actions and transition model



Distance in miles

# Example: Vacuum world

State Space



- **Initial State:** Defined by agent location and dirt location.
- **Actions:** Left, right, suck
- **Transition model and states**
  - There are 8 possible atomic states of the system.
  - Why is the number of states for  $n$  possible locations  $n(2^n)$ ?
- **Goal state:** All locations are clean.
- **Path cost:** E.g., number of actions



# Example: Sliding-tile puzzle

- **Initial State:** A given configuration.
- **Actions:** Move blank left, right, up, down
- **States as a result of the Initial state and the Transition model**
  - The location of each tile (including the empty one,  $\frac{1}{2}$  of the permutations are unreachable)
    - 8-puzzle:  $9!/2 = 181,440$  states
    - 15-puzzle:  $16!/2 \approx 10^{13}$  states
    - 24-puzzle:  $25!/2 \approx 10^{25}$  states
- **Goal state:** Tiles are arranged empty and 1-8
- **Path cost:** 1 per tile move.

|   |   |   |
|---|---|---|
| 7 | 2 | 4 |
| 5 |   | 6 |
| 8 | 3 | 1 |

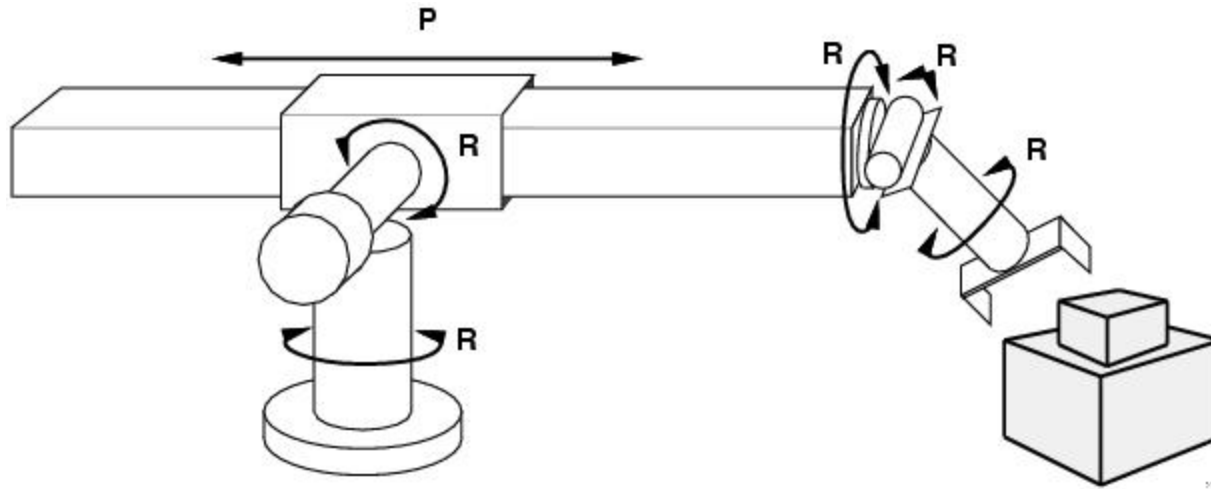
Start State

|   |   |   |
|---|---|---|
|   | 1 | 2 |
| 3 | 4 | 5 |
| 6 | 7 | 8 |

Goal State



# Example: Robot motion planning



- **Initial State:** Current arm position.
- **States:** Real-valued coordinates of robot joint angles.
- **Actions:** Continuous motions of robot joints.
- **Goal state:** Desired final configuration (e.g., object is grasped).
- **Path cost:** Time to execute, smoothness of path, etc.

# Solving search problems

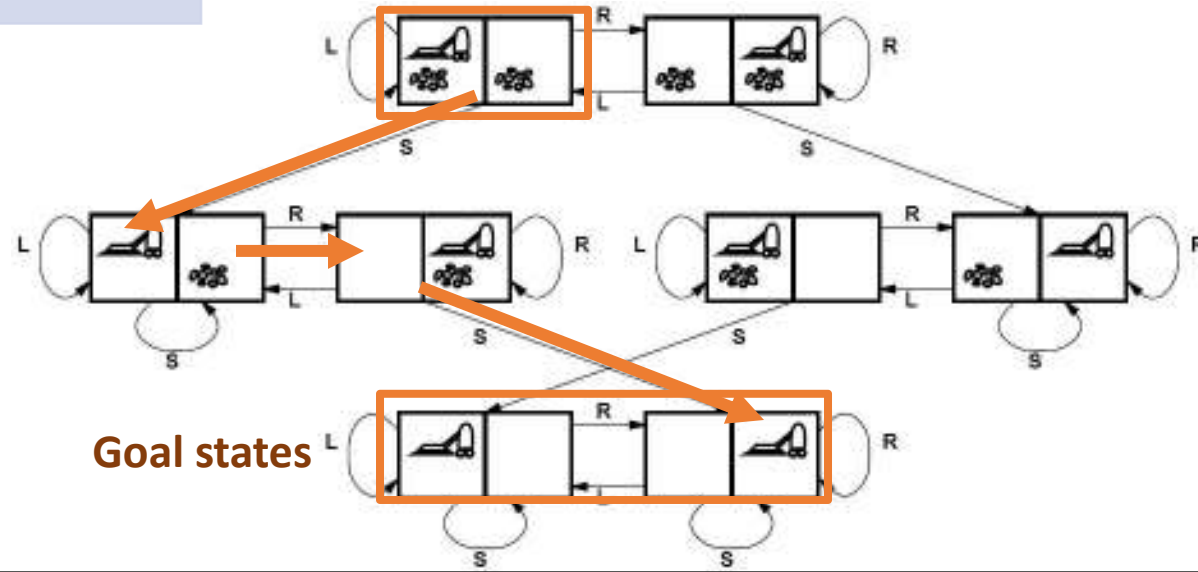
Given a search problem definition

- Initial state
- Actions
- Transition model
- Goal state
- Path cost

**How do we find the optimal solution (sequence of actions/states)?**

**State space**

**Initial state**



**Goal states**

# Solving search problems

Given a search problem definition

- Initial state
- Actions
- Transition model
- Goal state
- Path cost

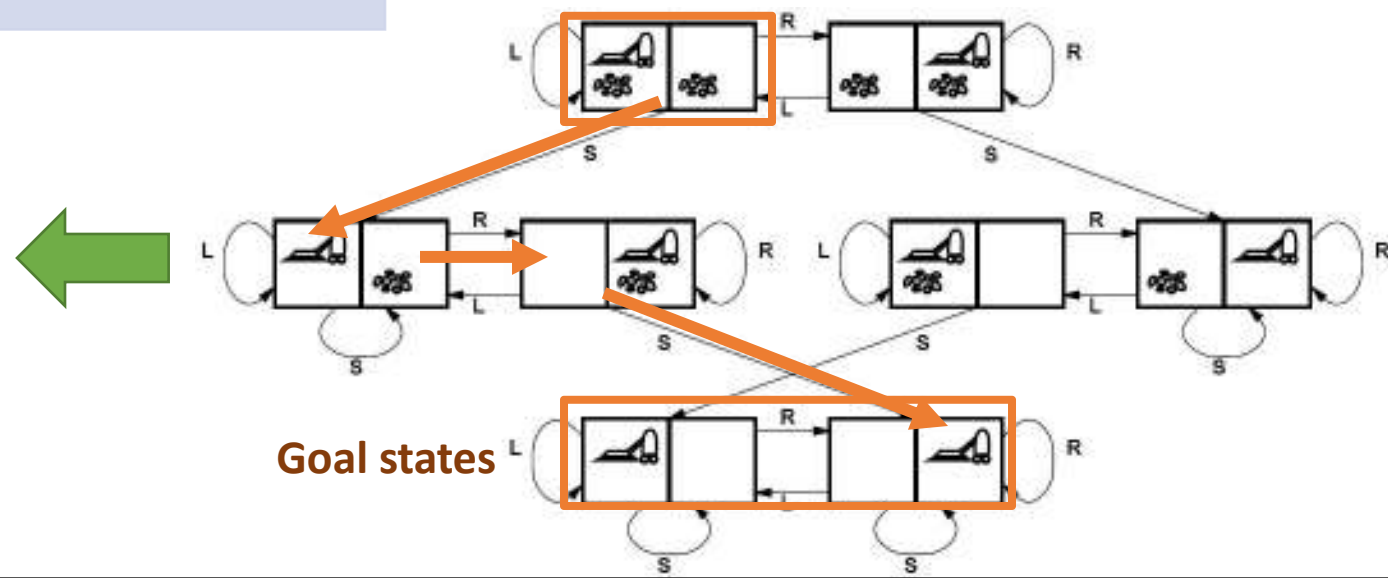
**How do we find the optimal solution (sequence of actions/states)?**

**State space**

**Initial state**

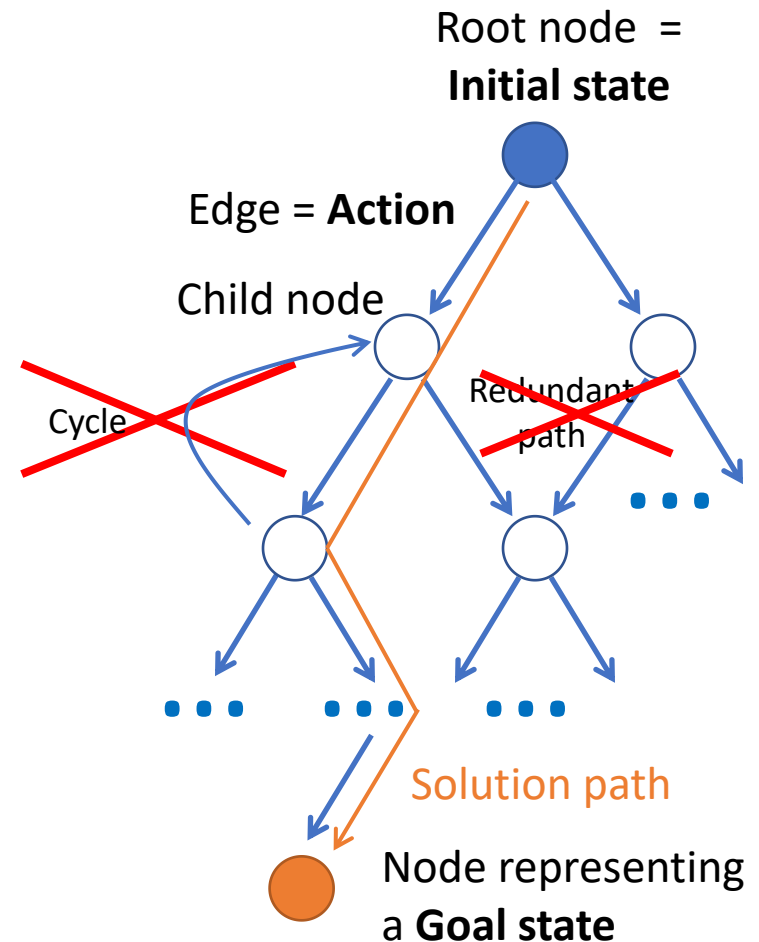
**Goal states**

Construct a search tree for the state space graph!



# Search tree

- Superimpose a “what if” tree of possible actions and outcomes (states) on the state space graph.
- The **Root node** represents the initial state.
- An action child node is reached by an **edge** representing an action. The corresponding state is defined by the transition model.
- Trees have no **cycles** or **redundant paths**. Cycles in the search space need to be broken. Removing redundant paths improves search efficiency.
- A **path** through the tree corresponds to a sequence of actions (states).
- A **solution** is a path ending in a node representing a goal state.
- **Nodes vs. states**: Each node represents a state of the environment. It contains the data structure that creates the search tree.



# Differences between typical Tree search and AI search

## Typical tree search

- Assumes a given tree that fits in memory.
- Trees have no cycles or redundant paths.

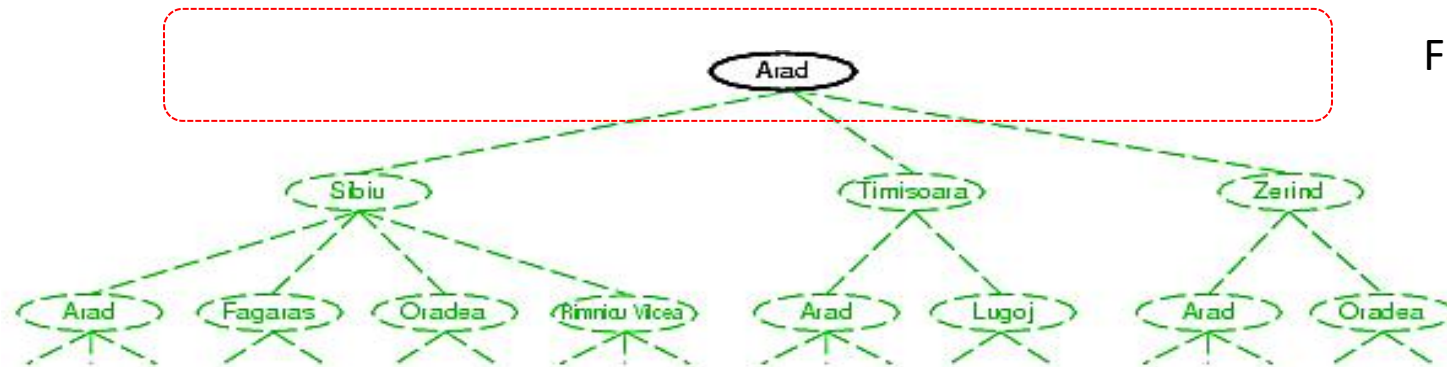
## AI tree/graph search

- The search space is too large to fit into **memory**.
  - a. **Builds parts of the tree** from initial state and transition function representing the graph.
  - b. **Memory management** is very important.
- The search space is typically very large and complicated graph. Memory-efficient **cycle checking** is very important to avoid infinite loops or minimize searching parts of the search space multiple times.
- Checking redundant paths often requires too much memory.

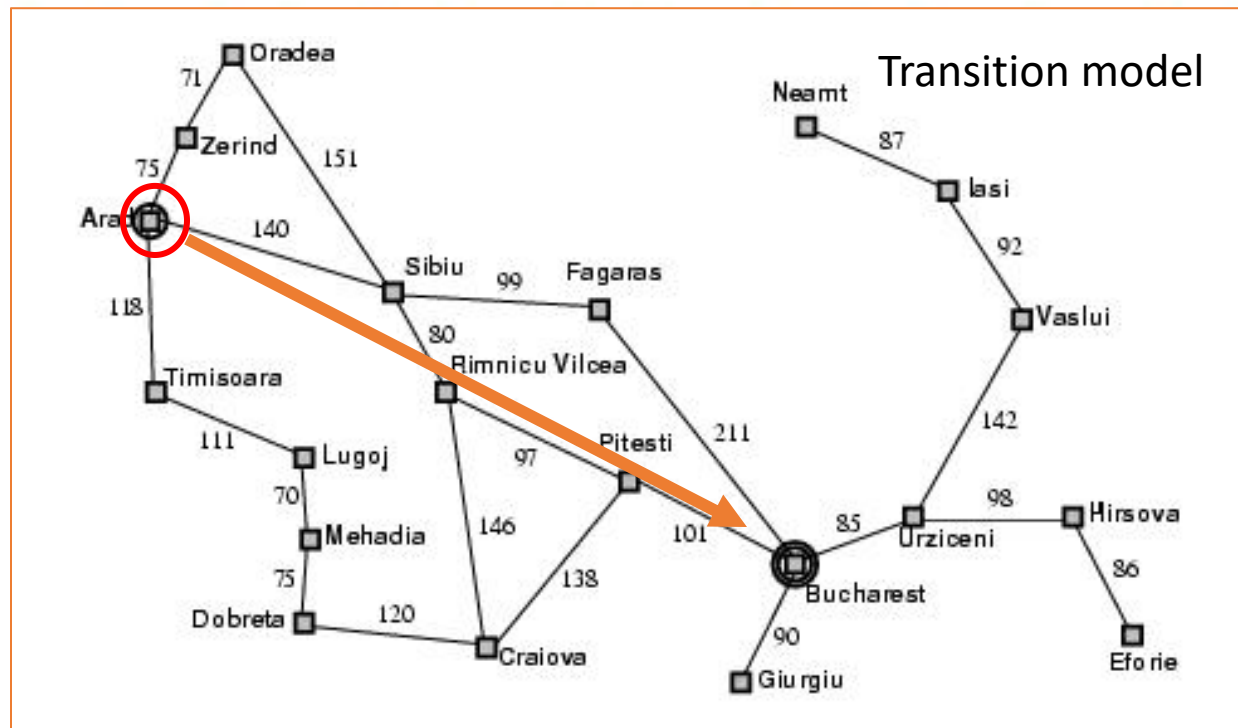
# Tree Search Algorithm Outline

1. Initialize the **frontier** (set of unexplored know nodes) using the **starting state/root node**.
2. While the frontier is not empty:
  - a) Choose a frontier node to expand according to **search strategy**.
  - b) If the node represents a **goal state**, return it as the solution.
  - c) Else **expand** the node (i.e., apply all possible actions to the transition model) and add its children nodes representing the newly reached states to the frontier.

# Tree search example

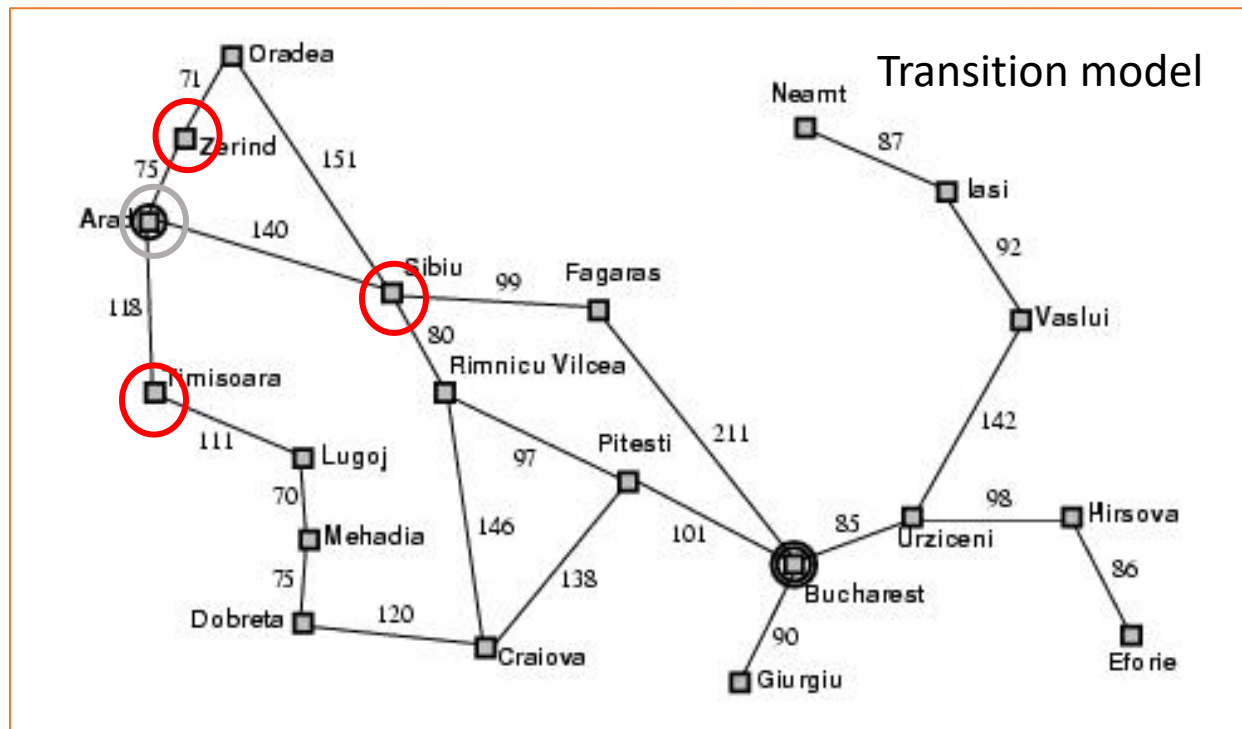
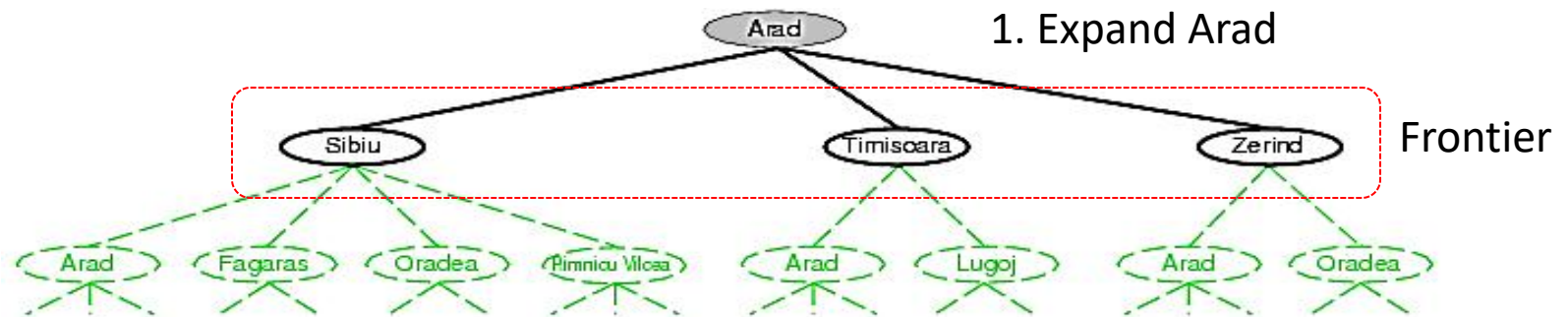


Frontier

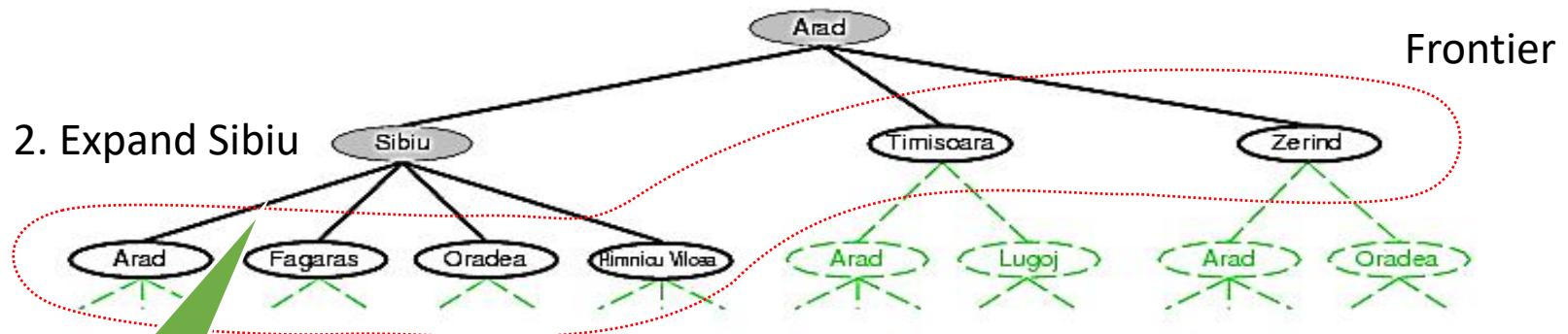




# Tree search example

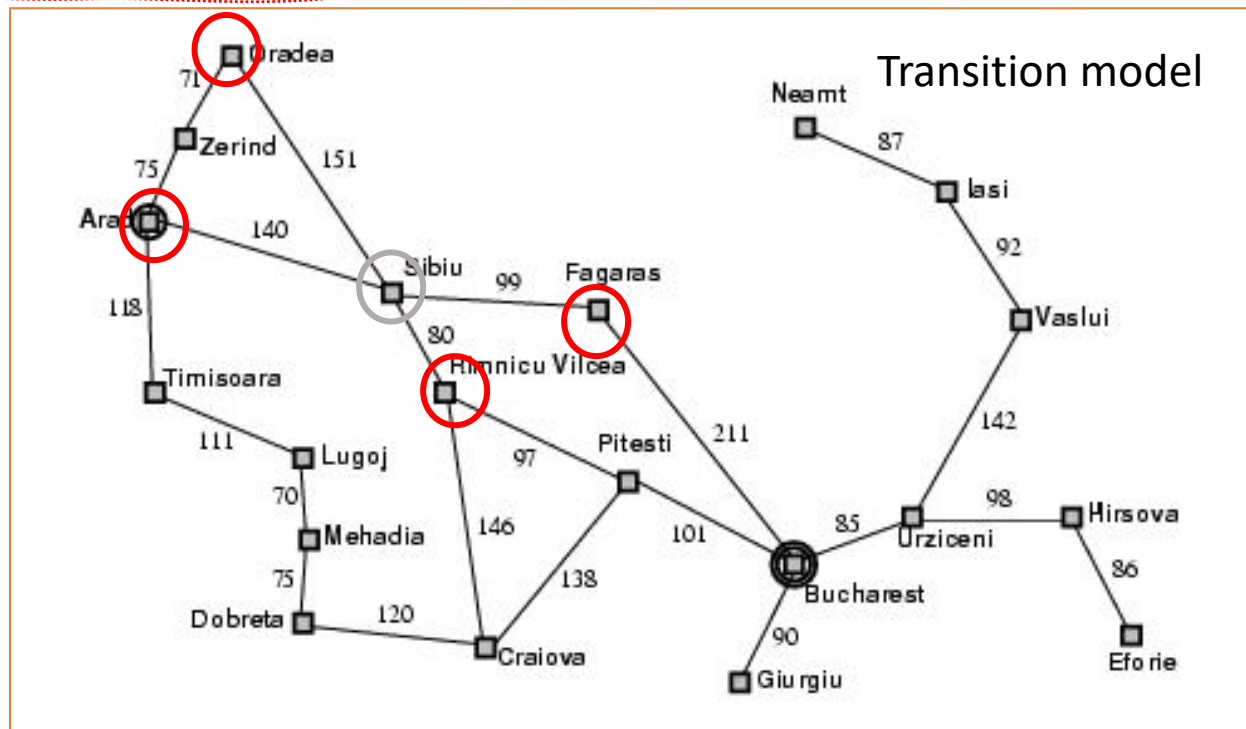


# Tree search example



Example of a  
redundant path

We could have  
also expanded  
Timisoara or  
Zerind!



# Search strategies

- A **search strategy** is defined by picking the **order of node expansion**.
- Strategies are evaluated along the following dimensions:
  - **Completeness**: does it always find a solution if one exists?
  - **Optimality**: does it always find a least-cost solution?
  - **Time complexity**: how long does it take?
  - **Space complexity**: how much memory does it need?
- Worst case time and space complexity are measured in terms of the **size of the state space  $n$**  (= number of nodes in the search tree).  
Metrics used if the state space is only implicitly defined by initial state, actions and a transition function are:
  - $d$ : depth of the optimal solution (= number of actions needed)
  - $m$ : the number of actions in any path (may be infinite with loops)
  - $b$ : maximum branching factor of the search tree (number of successor nodes for a parent)

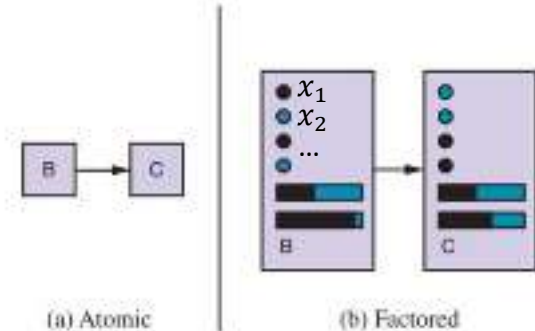


# State Space for Search

# State Space

- Number of different states the agent and environment can be in.
- **Reachable states** are defined by the initial state and the transition model. Not all states may be reachable from the initial state.
- **Search tree** spans the state space. Note that a single state can be represented by several search tree nodes if we have redundant paths.
- State space size is an indication of problem size.

## State representation



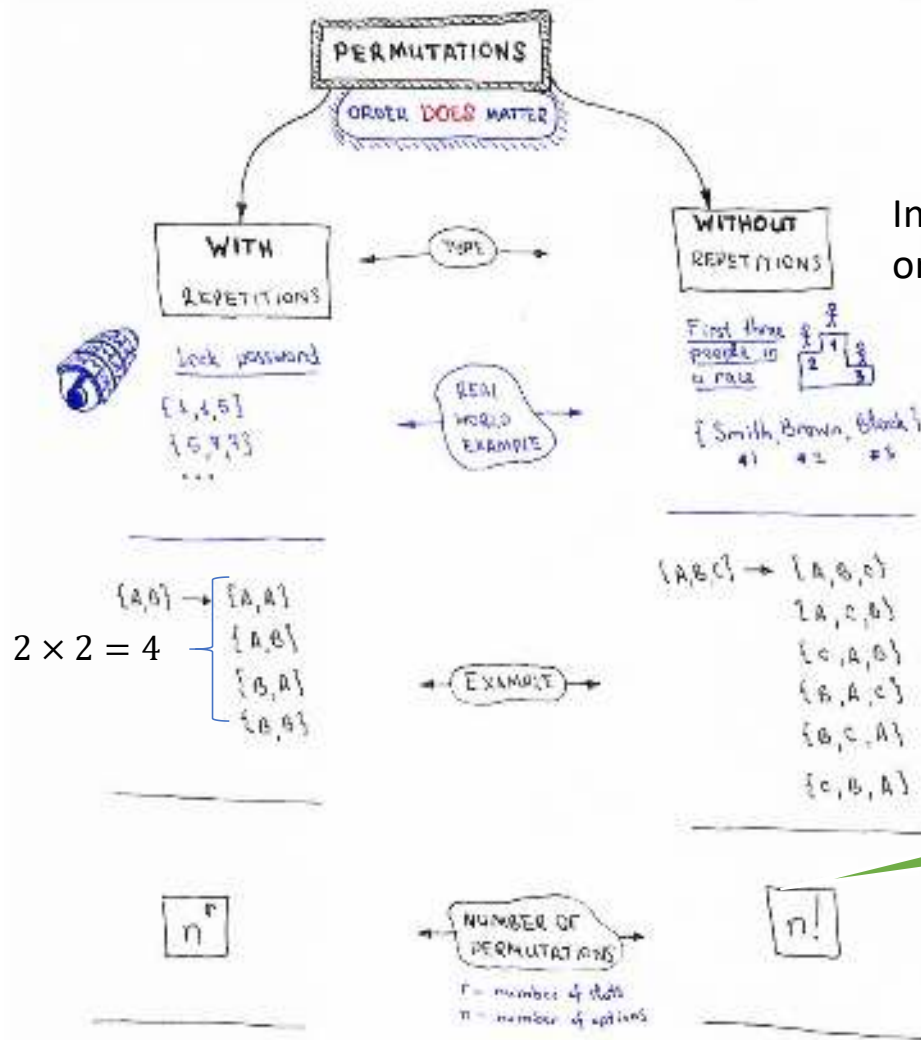
## State Space Size Estimation

- Even if the used algorithm represents the state space using atomic states, we may know that internally they are factored and can use this information to estimate the problem size.
- Basic rule to calculate (estimate) the state space size for factored state representation with  $n$  variables is:

$$|x_1| \times |x_2| \times \dots \times |x_n|$$

where  $|\cdot|$  is the number of possible values.



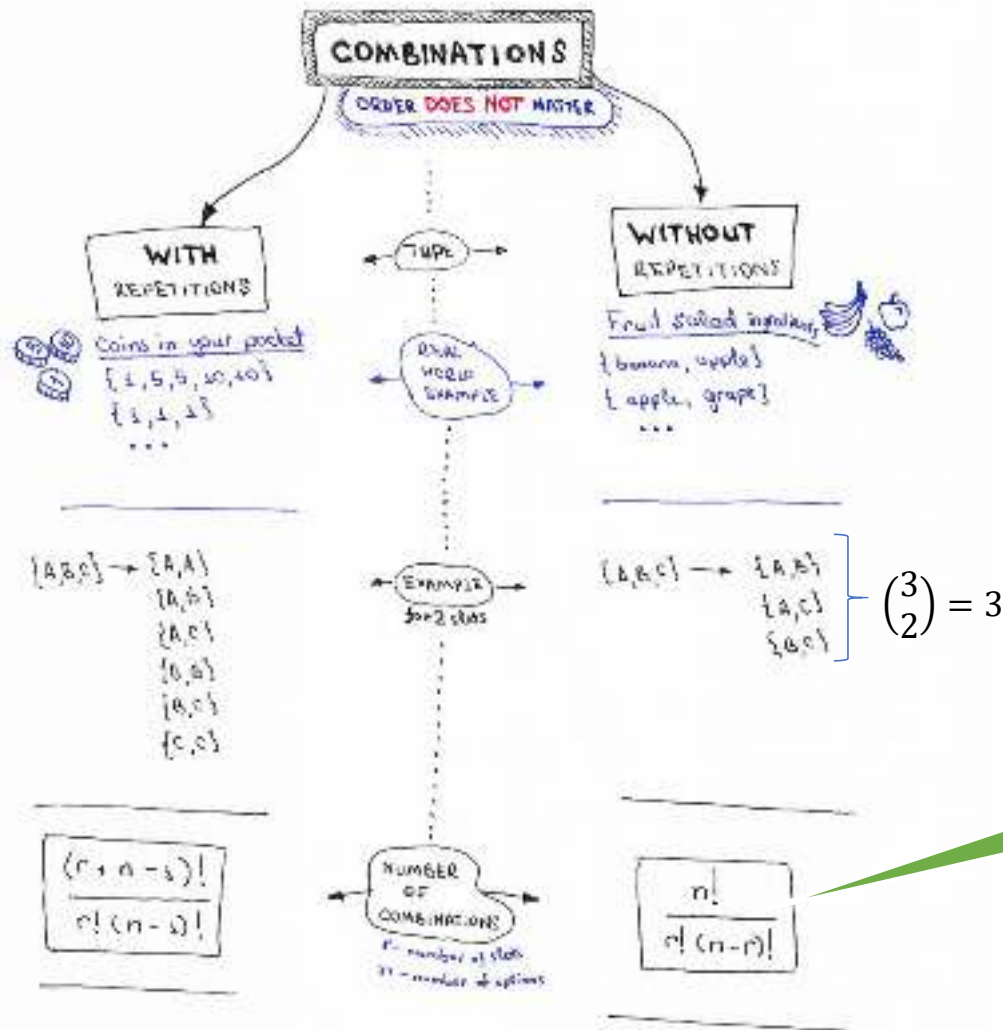


In how many ways can we order/arrange n objects?

**Factorial:**  $n! = n \times (n - 1) \times \dots \times 2 \times 1$

**#Python**  
**import math**

**print (math.factorial(23))**

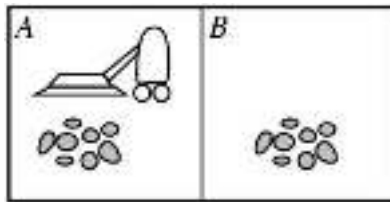


**Binomial Coefficient:**  $\binom{n}{r} = C(n, r) = {}_nC_r$   
Read as “n choose r” because it is the number of ways can we choose r out of n objects?  
Special case for  $r = 2$ :  $\binom{n}{2} = \frac{n(n-1)}{2}$

**#Python**  
`import scipy.special`

**# the two give the same results**  
`scipy.special.binom(10, 5)`  
`scipy.special.comb(10, 5)`

# Examples: What is the state space size?



$n$  = number of squares

## Dirt

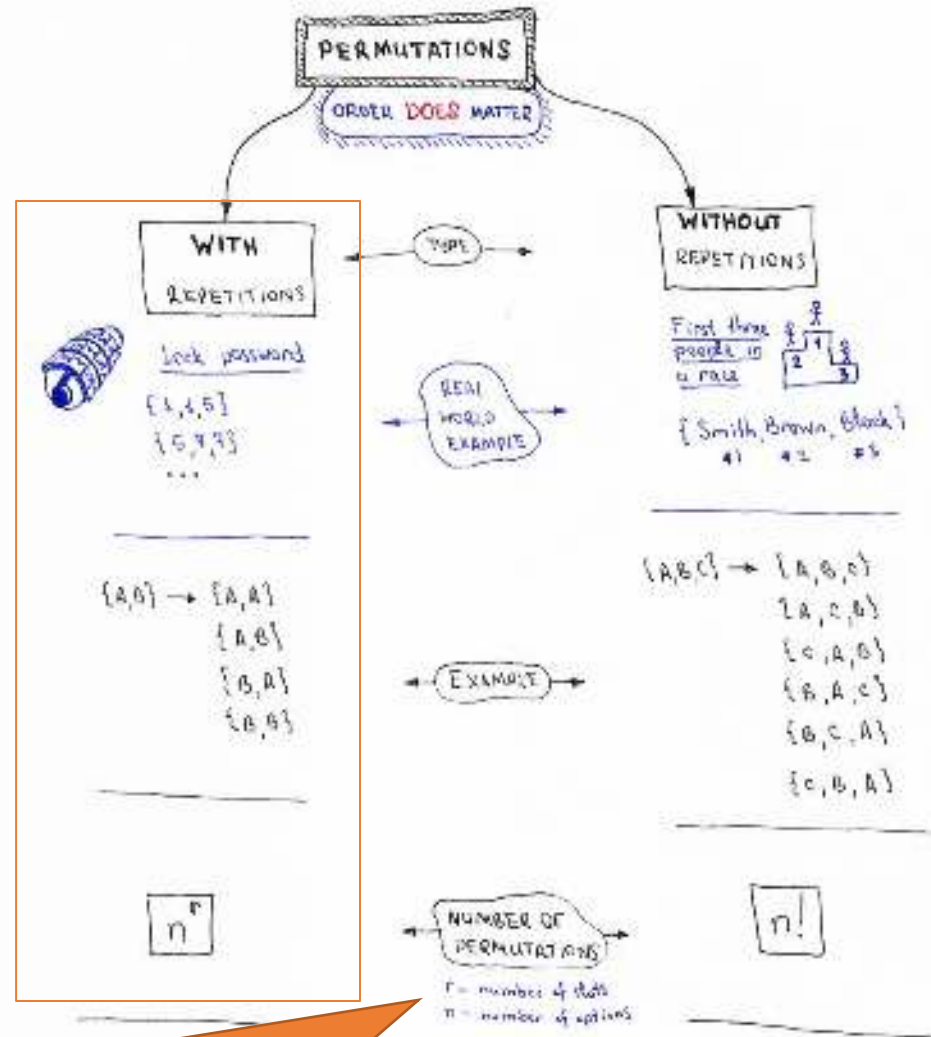
- Order of A and B does not matter!
- Repetition: Dirt can be in both places
- There are 2 options (clean/dirty)

$$\rightarrow 2^n$$

## Robot

- Can be in 1 out of  $n$  places.
- $$\rightarrow n$$

Total:  $n2^n$



We call the number of squares ("slots")  $n$  here!



# Examples: What is the state space size?

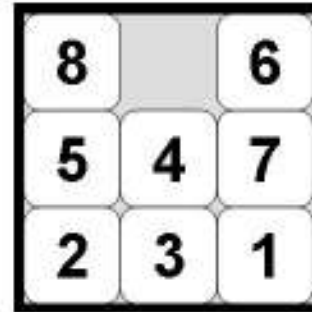
Often a rough upper limit is sufficient to determine how hard the search problem is.



Maze



8-queens problem



8-puzzle problem



Tic-tac-toe

# Uninformed Search



# Uninformed search strategies

The search algorithm/agent is **not** provided information about how close a state is to the goal state.

It blindly searches following a simple strategy until it finds the goal state by chance.

Search strategies we will discuss:

Breadth-first search

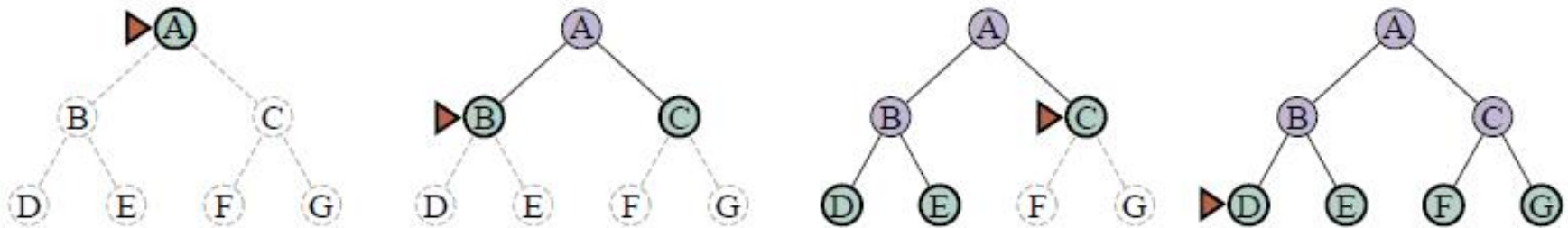
Uniform-cost search

Depth-first search

Iterative deepening search

# Breadth-first search (BFS)

**Expansion rule:** Expand shallowest unexpanded node in the frontier (=FIFO).



**Figure 3.8** Breadth-first search on a simple binary tree. At each stage, the node to be expanded next is indicated by the triangular marker.

## Data Structures

- **Frontier** data structure: holds references to the green nodes (green) and is implemented as a FIFO queue.
- **Reached** data structure: holds references to all visited nodes (gray and green) and is used to prevent visiting nodes more than once (cycle checking).
- Builds a **tree** with links from parent to child.

# Implementation: BFS

```
function BREADTH-FIRST-SEARCH(problem) returns a solution node or failure  
  node  $\leftarrow$  NODE(problem.INITIAL)  
  if problem.IS-GOAL(node.STATE) then return node  
  frontier  $\leftarrow$  a FIFO queue, with node as an element  
  reached  $\leftarrow$  {problem.INITIAL}  
  while not IS-EMPTY(frontier) do  
    node  $\leftarrow$  POP(frontier)  
    for each child in EXPAND(problem, node) do  
      s  $\leftarrow$  child.STATE  
      if problem.IS-GOAL(s) then return child  
      if s is not in reached then  
        add s to reached  
        add child to frontier  
  return failure
```

**Expand** adds the next level below node to the search tree.

**reached** makes sure we do not visit nodes twice (e.g., in a cycle or a redundant path). Fast lookup is important.

# Implementation: Expanding the search tree

- AI tree search creates the search tree while searching.
- The EXPAND function uses the current search tree node (i.e., current state) and the problem description to create new nodes for all reachable states.
- It tries all actions in the current state by checking the transition function (RESULTS) and then returns a list of new nodes for the frontier.

```
function EXPAND(problem, node) yields nodes
  s ← node.STATE
  for each action in problem.ACTIONS(s) do
    s' ← problem.RESULT(s, action)
    cost ← node.PATH-COST + problem.ACTION-COST(s, action, s')
    yield NODE(STATE=s', PARENT=node, ACTION=action, PATH-COST=cost)
```

Transition  
function

Node structure for  
the search tree.  
Yield can also be  
implemented by  
returning a list of  
Nodes.



# Properties of breadth-first search

- **Complete?**

Yes

$d$ : depth of the optimal solution  
 $m$ : max. depth of tree  
 $b$ : maximum branching factor

- **Optimal?**

Yes – if cost is the same per step (action). Otherwise: Use uniform-cost search.

- **Time?**

Sum of the number of nodes created in at each level in a  $b$ -ary tree of depth  $d$ :

$$1 + b + b^2 + \dots + b^d = O(b^d)$$

- **Space?**

Stored nodes:  $O(b^d)$

**Note:**

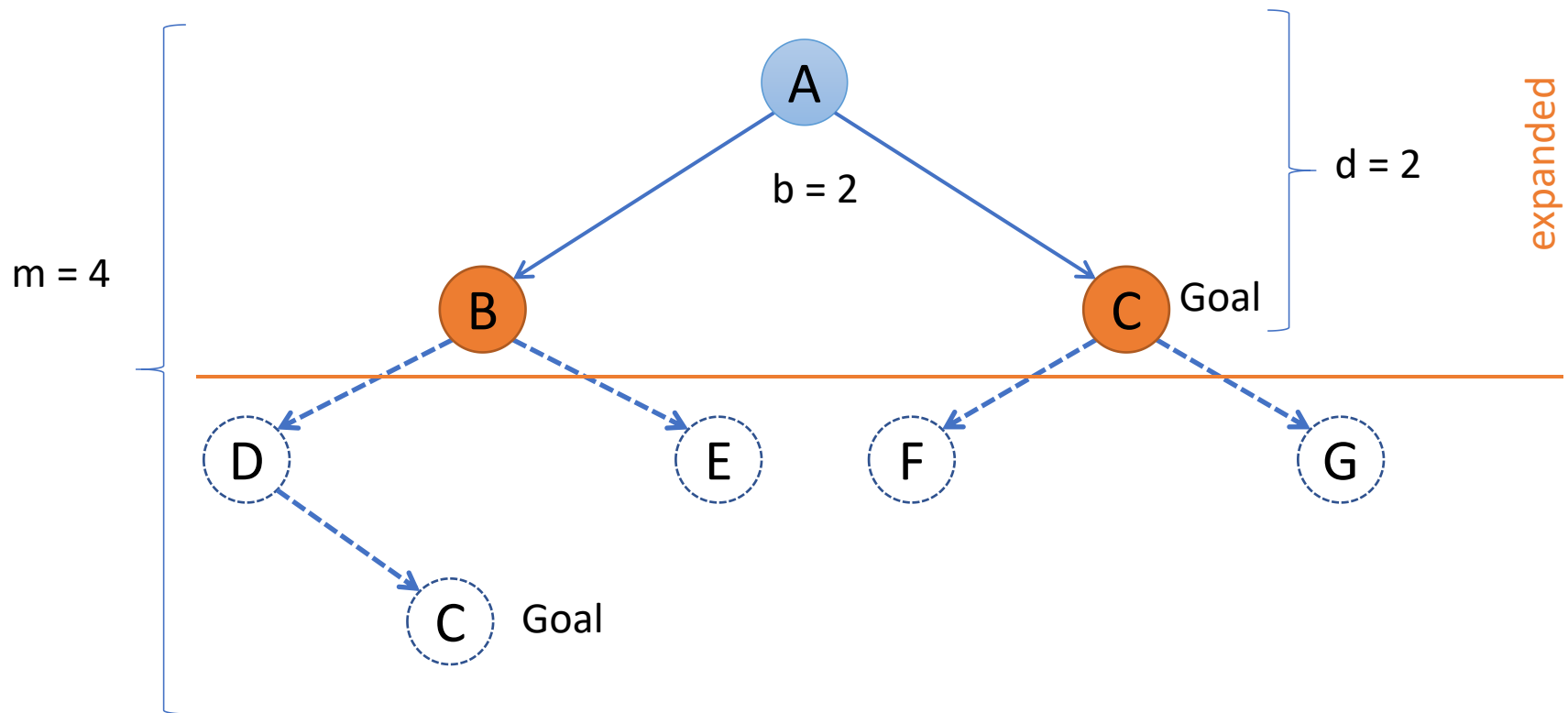
- The large space complexity is usually a bigger problem than time!



# Breadth-first search

$d$ : depth of the optimal solution  
 $m$ : max. depth of tree  
 $b$ : maximum branching factor

- Time and Space:  $O(b^d)$  - all paths to the depth of the goal are expanded



# Uniform-cost search

## (= Dijkstra's shortest path algorithm)

- **Expansion rule:** Expand node in the frontier with **the least path cost** from the initial state.
- Implementation: **best-first search** where the frontier is a priority queue ordered by  $f(n) = \text{path cost}$  from smallest to largest.
- Breadth-first search is a special case when all step costs being equal, i.e., each action costs the same!

- **Complete?**

Yes, if all step cost is greater than some small positive constant  $\epsilon > 0$

$d$ : depth of the optimal solution  
 $m$ : max. depth of tree  
 $b$ : maximum branching factor

- **Optimal?**

Yes – nodes expanded in increasing order of path cost

- **Time?**

Number of nodes with path cost  $\leq$  cost of optimal solution ( $C^*$ ) is  $O(b^{1+C^*/\epsilon})$ .

This can be greater than  $O(b^d)$ : the search can explore long paths consisting of small steps before exploring shorter paths consisting of larger steps

- **Space?**

$O(b^{1+C^*/\epsilon})$

See [Dijkstra's algorithm on Wikipedia](#)

# Implementation: Best-First Search Strategy

```
function UNIFORM-COST-SEARCH(problem) returns a solution node, or failure  
return BEST-FIRST-SEARCH(problem, PATH-COST)
```

```
function BEST-FIRST-SEARCH(problem, f) returns a solution node or failure  
  node  $\leftarrow$  NODE(STATE=problem.INITIAL)  
  frontier  $\leftarrow$  a priority queue ordered by f, with node as an element  
  reached  $\leftarrow$  a lookup table, with one entry with key problem.INITIAL and value node  
  while not IS-EMPTY(frontier) do  
    node  $\leftarrow$  POP(frontier)  
    if problem.IS-GOAL(node.STATE) then return node  
    for each child in EXPAND(problem, node) do  
      s  $\leftarrow$  child.STATE  
      if s is not in reached or child.PATH-COST < reached[s].PATH-COST then  
        reached[s]  $\leftarrow$  child  
        add child to frontier  
  return failure
```

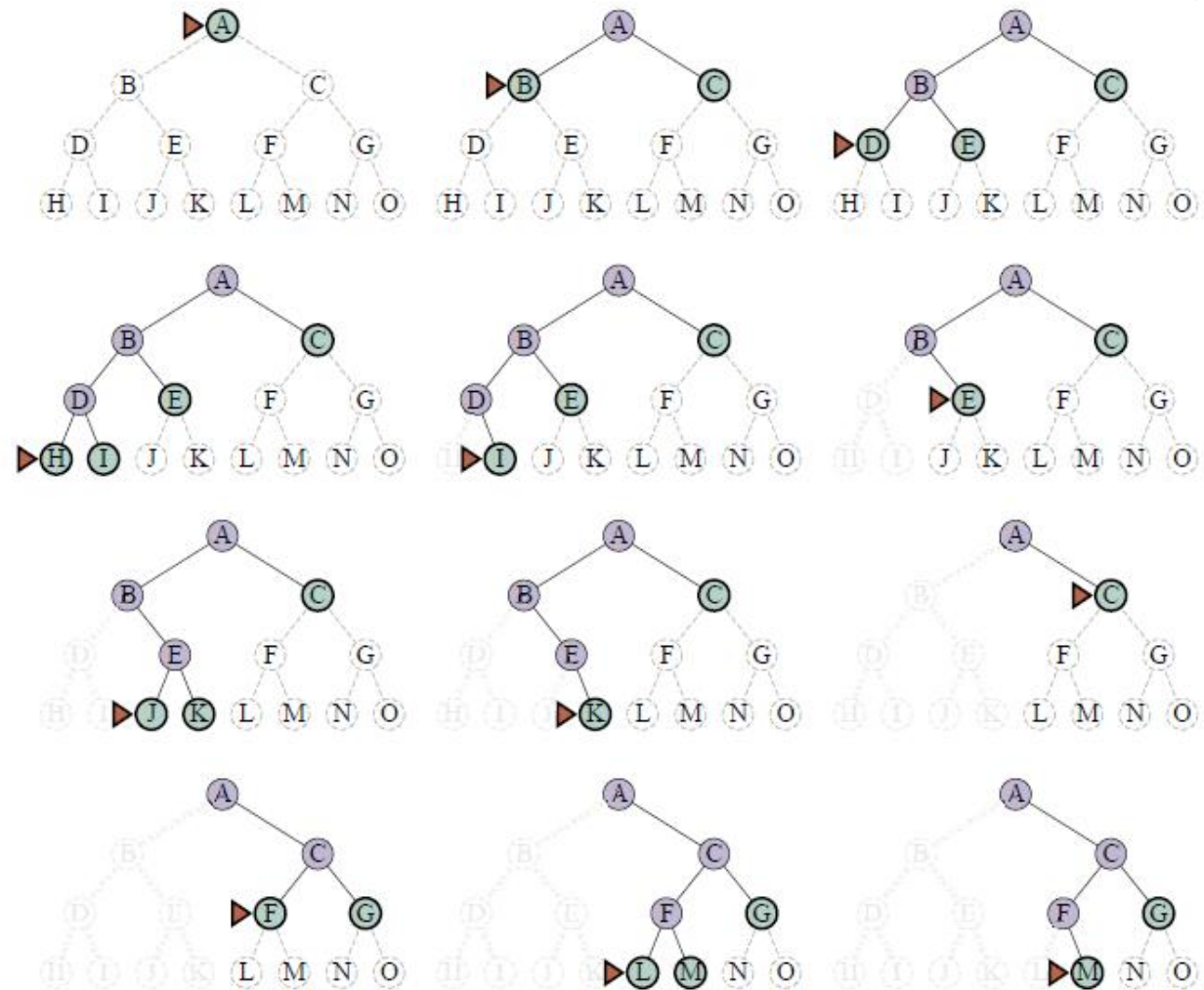
The order for expanding the frontier is determined by  $f(n)$  = path cost to node  $n$ .

This check is the difference to BFS! It visits a node again if it can be reached by a better (cheaper) path.

See BFS for function EXPAND.

# Depth-first search (DFS)

- **Expansion rule:** Expand deepest unexpanded node in the frontier (last added).
- **Frontier:** stack (LIFO)
- **Cycle Checking:** **No reached data structure!** Cycle checking can only check the current path and the frontier. This may lead to infinite loops!



**Figure 3.11** A dozen steps (left to right, top to bottom) in the progress of a depth-first search on a binary tree from start state A to goal M. The frontier is in green, with a triangle marking the node to be expanded next. Previously expanded nodes are lavender, and potential future nodes have faint dashed lines. Expanded nodes with no descendants in the frontier (very faint lines) can be discarded.

# Implementation: DFS

- DFS could be implemented like BFS/Best-first search and just taking the last element from the frontier (LIFO).
- However, to reduce the space complexity to  $O(bm)$ , the reached data structure needs to be removed! Options:
  - Recursive implementation (cycle checking is a problem).
  - We can use a search tree where the abandoned branches are removed from memory (cycle checking is only done against the current path). This is similar to Backtracking search.

DFS uses  $\ell = \infty$

```

function DEPTH-LIMITED-SEARCH(problem,  $\ell$ ) returns a node or failure or cutoff
  frontier  $\leftarrow$  a LIFO queue (stack) with NODE(problem.INITIAL) as an element
  result  $\leftarrow$  failure
  while not IS-EMPTY(frontier) do
    node  $\leftarrow$  POP(frontier)
    if problem.IS-GOAL(node.STATE) then return node
    if DEPTH(node) >  $\ell$  then
      result  $\leftarrow$  cutoff
    else if not IS-CYCLE(node) do
      for each child in EXPAND(problem, node) do
        add child to frontier
  return result
  
```

If we only keep the current path in memory, then we can only check against the path from the root to the current node and the frontier to prevent cycles.

See BFS for function EXPAND.

# Properties of depth-first search

- **Complete?**

- Only in finite search spaces. Some cycles can be avoided by checking for repeated states along the path.
- **Incomplete in infinite search spaces** (e.g., with cycles).

- **Optimal?**

No – returns the first solution it finds.

$d$ : depth of the optimal solution  
 $m$ : max. depth of tree  
 $b$ : maximum branching factor

- **Time?**

Could be the time to reach a solution at maximum depth  $m$  in the last path:  $O(b^m)$   
Terrible if  $m \gg d$ , but if there are many shallow solutions, it can be much faster than BFS.

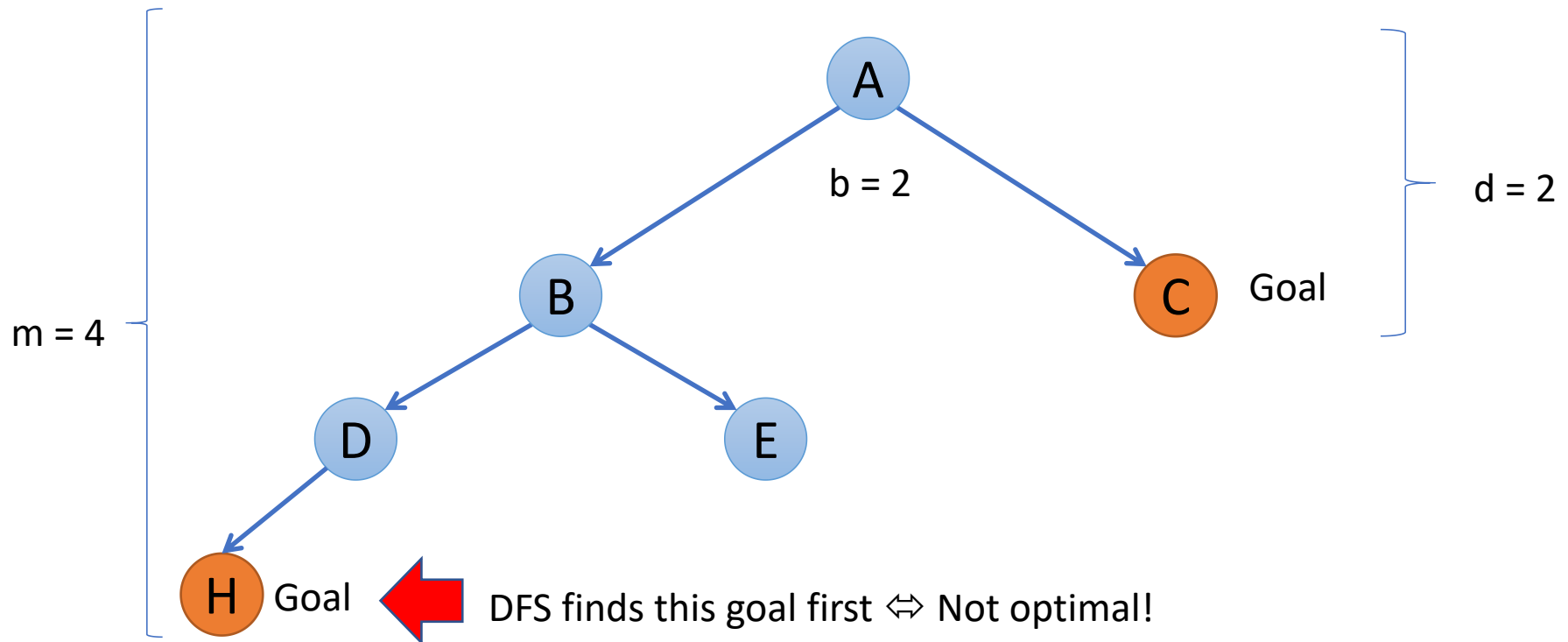
- **Space?**

$O(bm) \Leftrightarrow$  **linear space!** (if no reached data structure is used)

# Depth-first search

$d$ : depth of the optimal solution  
 $m$ : max. depth of tree  
 $b$ : maximum branching factor

- Time:  $O(b^m)$  – worst case is expanding all paths.
- Space:  $O(bm)$  - if it only stores the frontier nodes and the current path.



**Note:** The order in which we add new nodes to the frontier can change what goal we find!



# Iterative deepening search (IDS)

Can we

- a) get DFS's good memory footprint,**
- b) avoid infinite cycles, and**
- c) preserve BFS's optimality guaranty?**

Use depth-restricted DFS and gradually increase the depth.

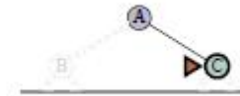
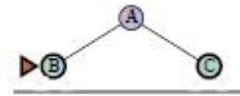
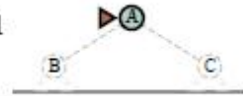
1. Check if the root node is the goal.
2. Do a DFS searching for a path of length 1
3. If goal not found, do a DFS searching for a path of length 2
4. If goal not found, do a DFS searching for a path of length 3
5. ...

# Iterative deepening search (IDS)

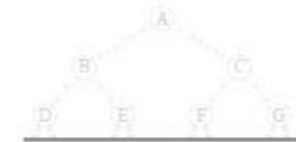
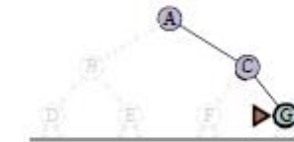
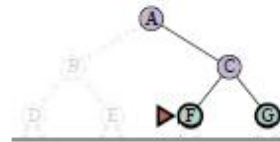
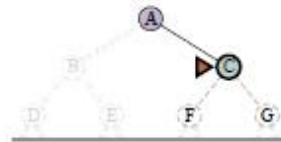
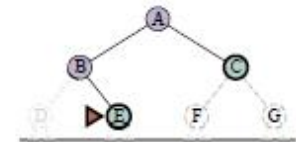
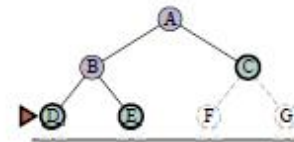
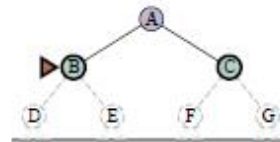
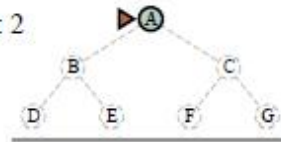
limit: 0



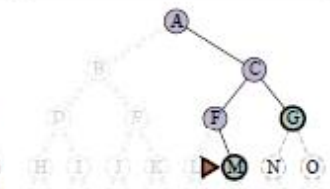
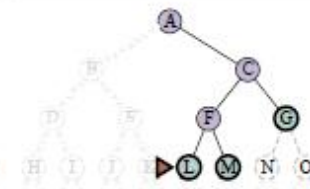
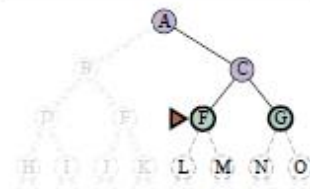
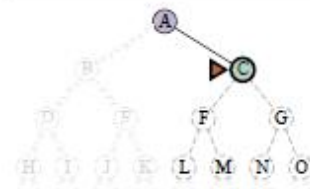
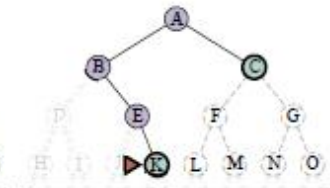
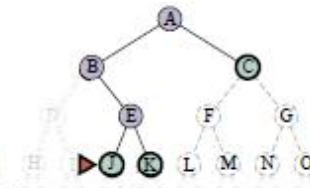
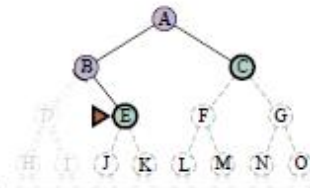
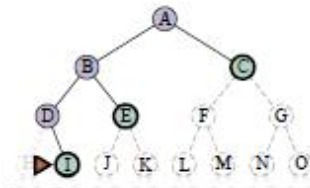
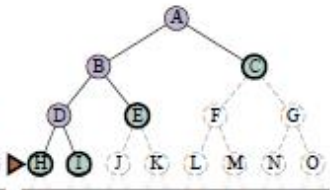
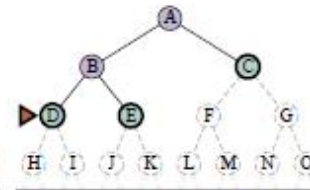
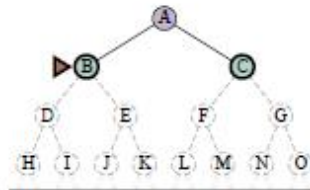
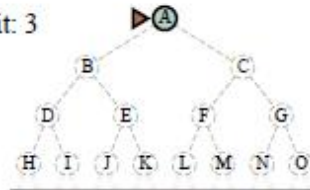
limit: 1



limit: 2



limit: 3



# Implementation: IDS

```
function ITERATIVE-DEEPENING-SEARCH(problem) returns a solution node or failure  
  for depth = 0 to  $\infty$  do  
    result  $\leftarrow$  DEPTH-LIMITED-SEARCH(problem, depth)  
    if result  $\neq$  cutoff then return result
```

```
function DEPTH-LIMITED-SEARCH(problem,  $\ell$ ) returns a node or failure or cutoff  
  frontier  $\leftarrow$  a LIFO queue (stack) with NODE(problem.INITIAL) as an element  
  result  $\leftarrow$  failure  
  while not IS-EMPTY(frontier) do  
    node  $\leftarrow$  POP(frontier)  
    if problem.IS-GOAL(node.STATE) then return node  
    if DEPTH(node) >  $\ell$  then  
      result  $\leftarrow$  cutoff  
    else if not IS-CYCLE(node) do  
      for each child in EXPAND(problem, node) do  
        add child to frontier  
  return result
```

See BFS for function EXPAND.

# Properties of iterative deepening search

- **Complete?**

Yes

$d$ : depth of the optimal solution  
 $m$ : max. depth of tree  
 $b$ : maximum branching factor

- **Optimal?**

Yes, if step cost = 1

- **Time?**

Consists of rebuilding trees up to  $d$  times

$d b^1 + (d - 1)b^2 + \dots + 1b^d = O(b^d) \Leftrightarrow$  Slower than BFS, but the same complexity!

- **Space?**

$O(bd) \Leftrightarrow$  linear space. Even less than DFS since  $m \leq d$ . Cycles need to be handled by the depth-limited DFS implementation.

**Note:** IDS produces the same result as BFS but trades better space complexity for worse run time.

This makes DFS into the  
workhorse of AI.

# Informed Search

# Informed search

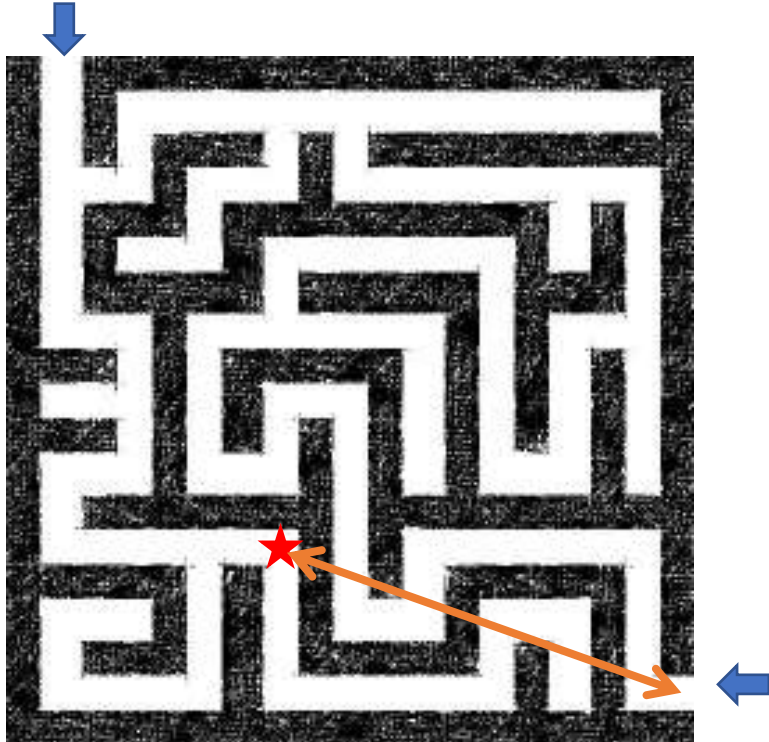
- AI search problems are typically very large. We would like to improve efficiency by **expanding as few nodes as possible**.
- The agent can use **additional information** in the form of “hints” about how promising different states/nodes are to lead to the goal. These hints are derived from
  - information the agent has (e.g., a map) or
  - percepts coming from a sensor.
- The agent uses a **heuristic function  $h(n)$**  to rank nodes in the frontier and select the most promising state in the frontier for expansion using a **best-first search** strategy.
- Algorithms:
  - Greedy best-first search
  - A\* search

# Heuristic function

- **Heuristic function**  $h(n)$  estimates the cost of reaching a node representing the goal state from the current node  $n$ .
- Examples:

**Euclidean distance**

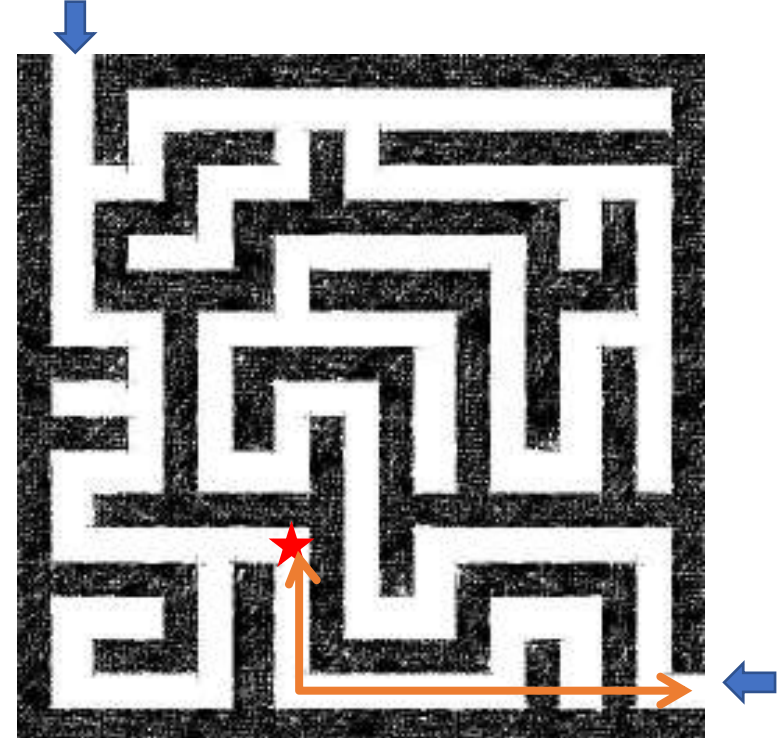
Start state



Goal state

**Manhattan distance**

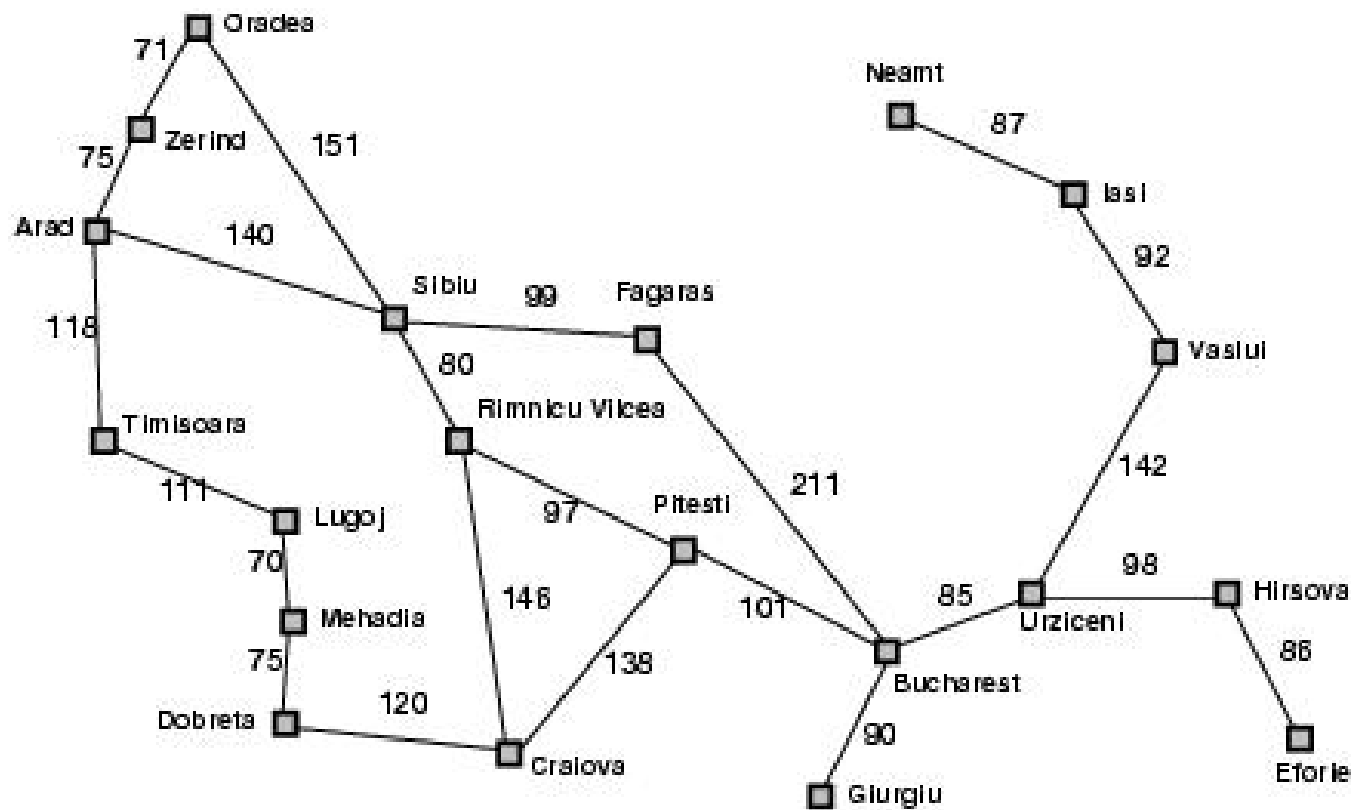
Start state



Goal state



# Heuristic for the Romania problem

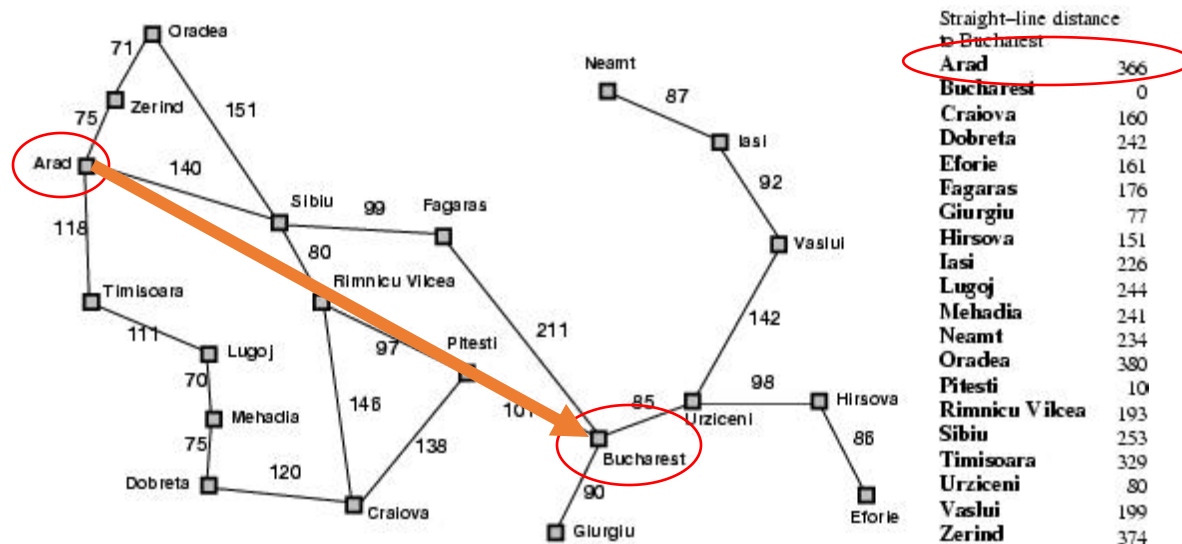


| Straight-line distance<br>to Bucharest | $h(n)$ |
|----------------------------------------|--------|
| Arad                                   | 366    |
| Bucharest                              | 0      |
| Craiova                                | 160    |
| Dobreta                                | 242    |
| Eforie                                 | 161    |
| Fagaras                                | 176    |
| Giurgiu                                | 77     |
| Hirsova                                | 151    |
| Iasi                                   | 226    |
| Lugoj                                  | 244    |
| Mehadia                                | 241    |
| Neamt                                  | 234    |
| Oradea                                 | 380    |
| Pitesti                                | 10     |
| Rimnicu Vilcea                         | 193    |
| Sibiu                                  | 253    |
| Timisoara                              | 329    |
| Urziceni                               | 80     |
| Vaslui                                 | 199    |
| Zerind                                 | 374    |

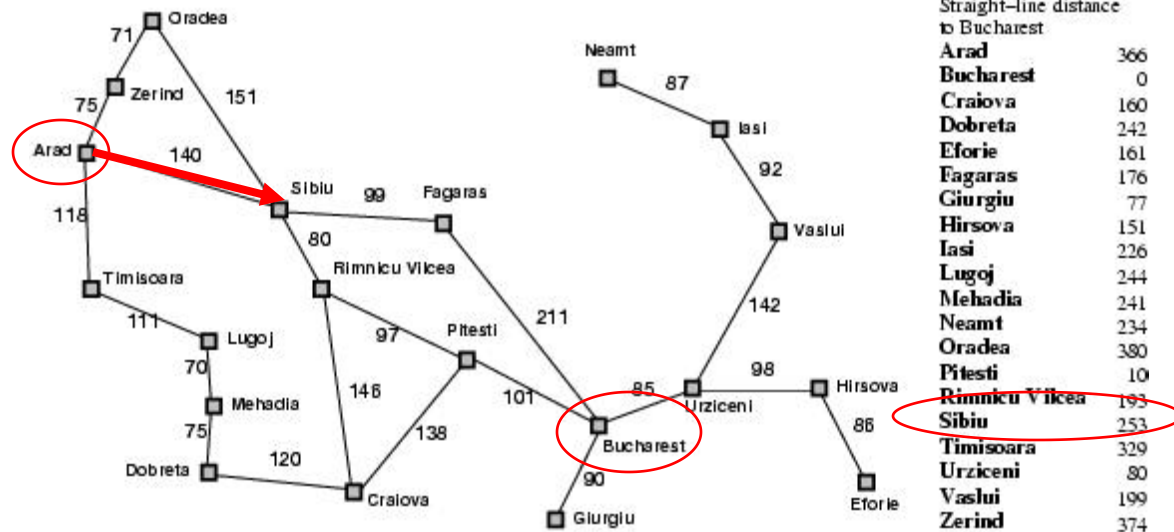
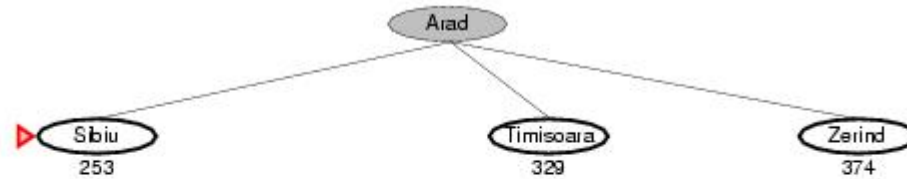
# Greedy best-first search example

Expansion rule: Expand the node that has the lowest value of the heuristic function  $h(n)$

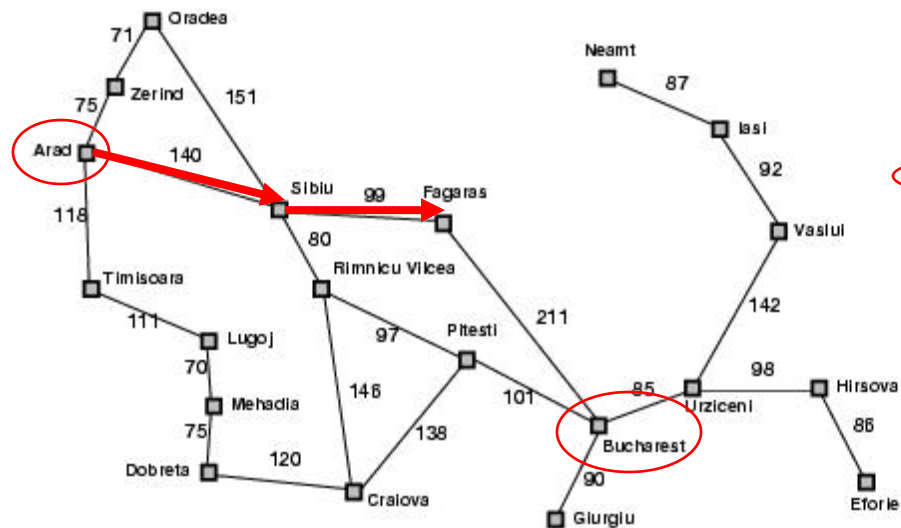
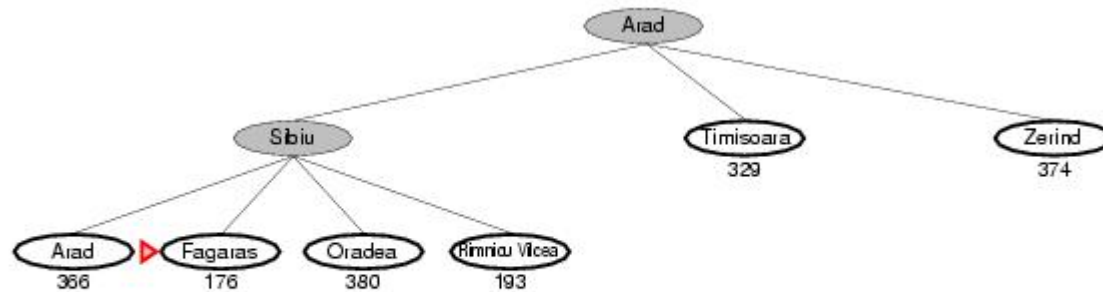
$$h(n) = \text{Arad} = 366$$



# Greedy best-first search example



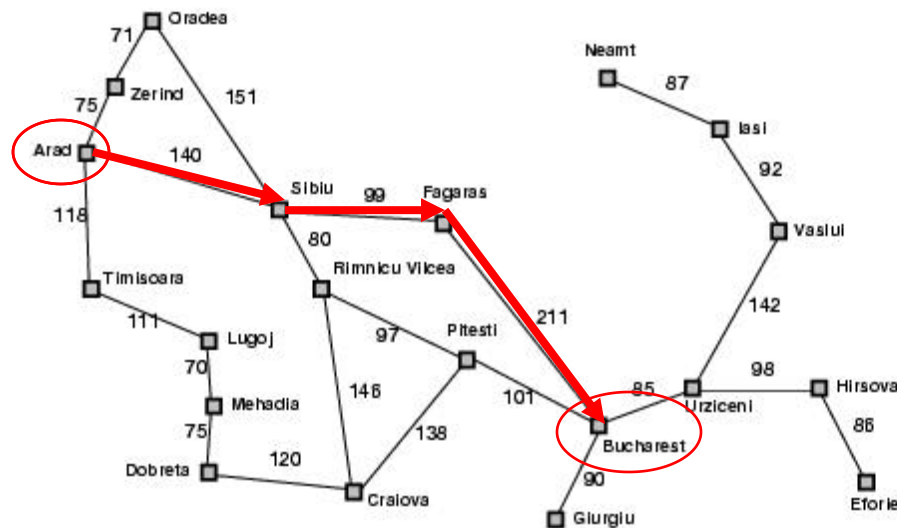
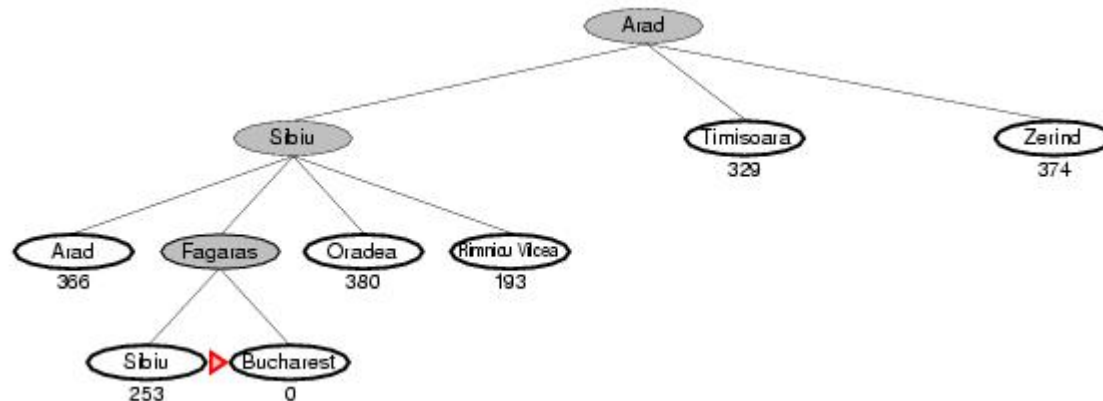
# Greedy best-first search example



Straight-line distance  
to Bucharest

|                |     |
|----------------|-----|
| Arad           | 366 |
| Bucharest      | 0   |
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| Dobreta        | 242 |
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# Greedy best-first search example



Straight-line distance  
to Bucharest

|                |     |
|----------------|-----|
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| Timisoara      | 329 |
| Urziceni       | 80  |
| Vaslui         | 199 |
| Zerind         | 374 |

Total:

$$140 + 99 + 211 = 450 \text{ miles}$$

# Properties of greedy best-first search

- **Complete?**

Yes – Best-first search is complete in finite spaces.

- **Optimal?**

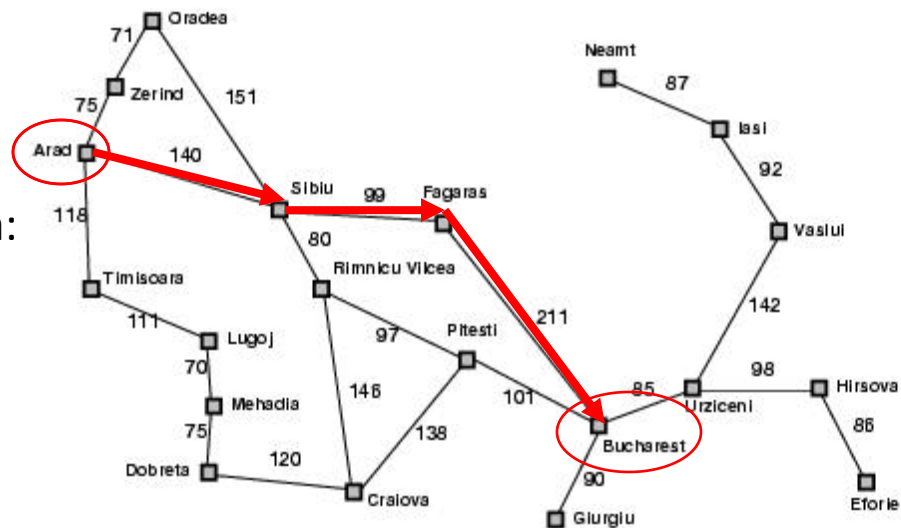
No

Total:

$$140 + 99 + 211 = 450 \text{ miles}$$

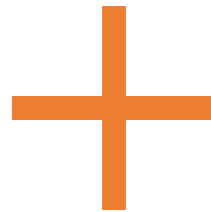
Alternative through Rimnicu Vilcea:

$$140 + 80 + 97 + 101 = 418 \text{ miles}$$



# Implementation of greedy best-first search

Best-First  
Search



Expand the frontier  
using  
 $f(n) = h(n)$



# Implementation of greedy best-first search

Heuristic  $h(n)$

```

function BEST-FIRST-SEARCH(problem, f) returns a solution node or failure
  node  $\leftarrow$  NODE(STATE=problem.INITIAL)
  frontier  $\leftarrow$  a priority queue ordered by f, with node as an element
  reached  $\leftarrow$  a lookup table, with one entry with key problem.INITIAL and value node
  while not IS-EMPTY(frontier) do
    node  $\leftarrow$  POP(frontier)
    if problem.IS-GOAL(node.STATE) then return node
    for each child in EXPAND(problem, node) do
      s  $\leftarrow$  child.STATE
      if s is not in reached or child.PATH-COST < reached[s].PATH-COST then
        reached[s]  $\leftarrow$  child
        add child to frontier
  return failure
  
```

The order for expanding the frontier is determined by  $f(n)$

This check is the different to BFS! It visits a node again if it can be reached by a better (cheaper) path.

See BFS for function EXPAND.

# Properties of greedy best-first search

- **Complete?**

Yes – Best-first search is complete in finite spaces.

- **Optimal?**

No

*d*: depth of the optimal solution  
*m*: max. depth of tree  
*b*: maximum branching factor

- **Time?**

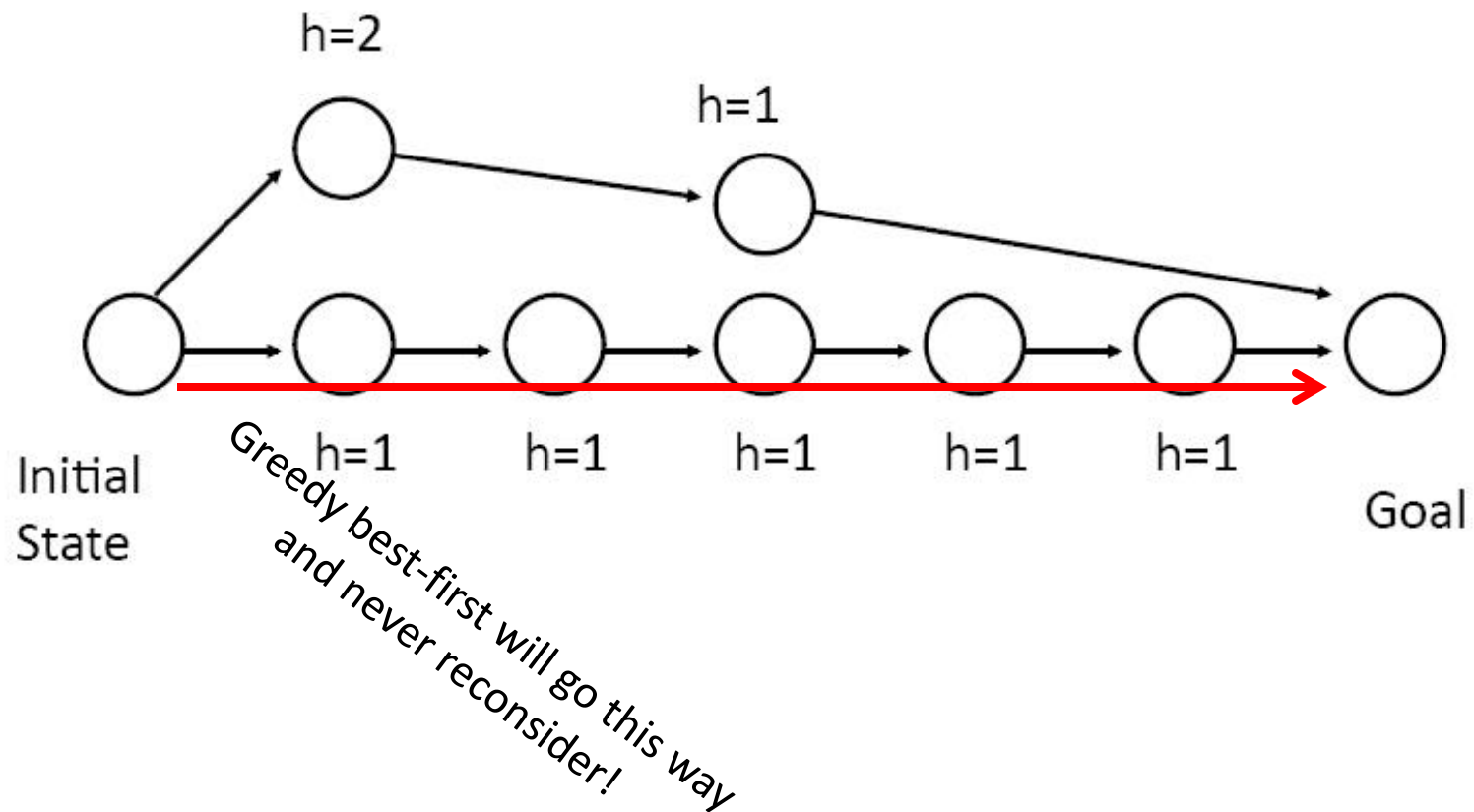
Worst case:  $O(b^m) \Leftrightarrow$  like DFS

Best case:  $O(bm)$  – If  $h(n)$  is 100% accurate

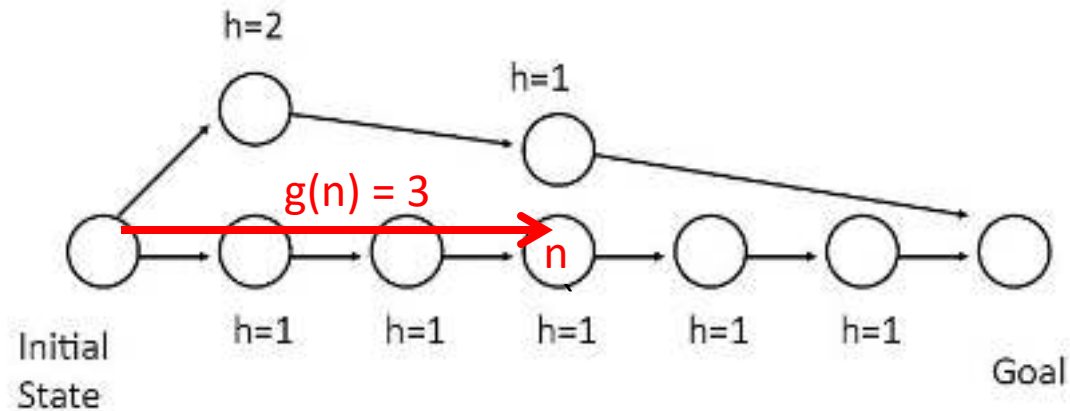
- **Space?**

Same as time complexity.

How can we fix the optimality problem with greedy best-first search?



# A\* search



- **Idea:** Take current path cost into account and avoid further expanding paths that are already very expensive.
- The evaluation function  $f(n)$  is the estimated total cost of the path through node  $n$  to the goal:

$$f(n) = g(n) + h(n)$$

$g(n)$ : cost so far to reach  $n$  (path cost)

$h(n)$ : estimated cost from  $n$  to goal (heuristic)

**Note:** For greedy best-first search we just use  $f(n) = h(n)$ .

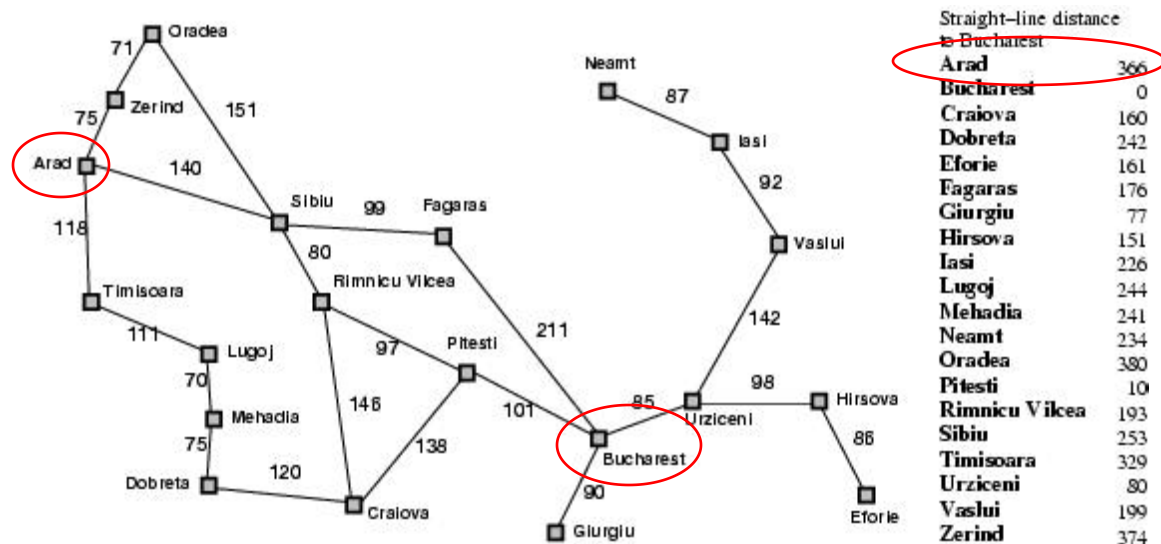
# A\* search example

Expansion rule:

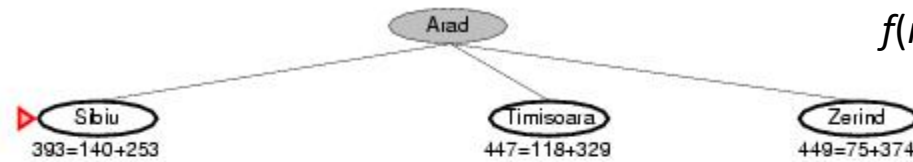
Expand the node with the smallest  $f(n)$

$$f(n) = g(n) + h(n) = 366 = 0 + 366$$

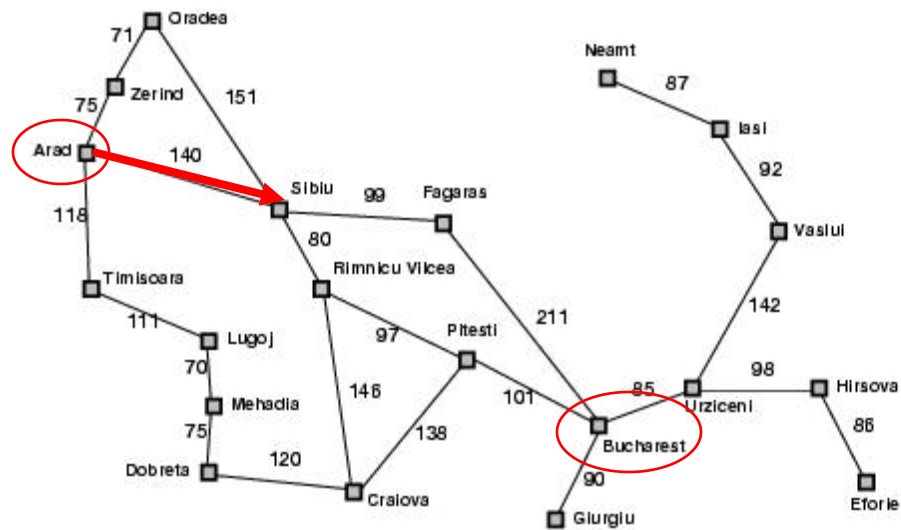
Arad



# A\* search example



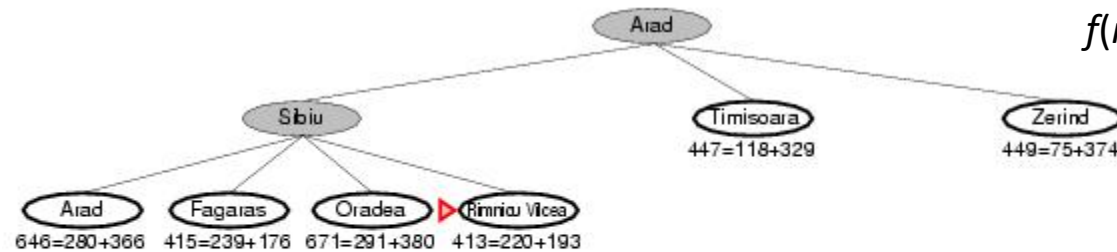
$$f(n) = g(n) + h(n)$$



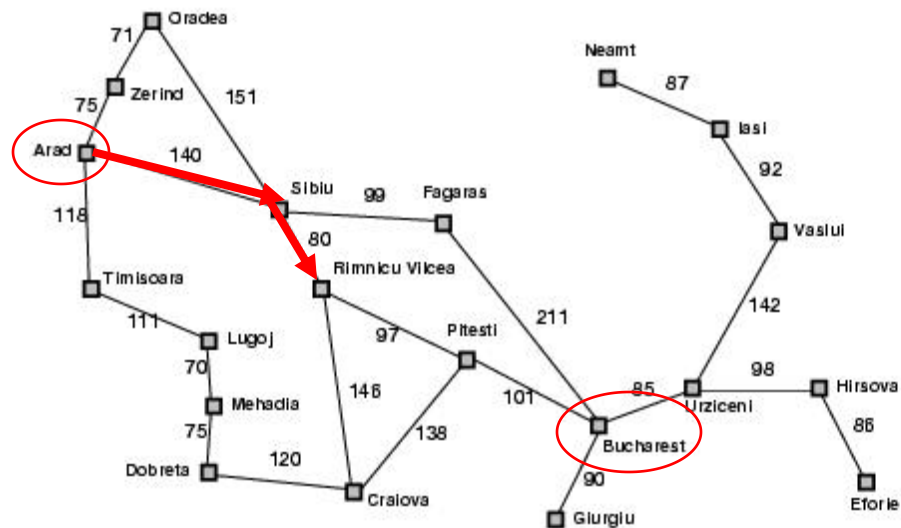
Straight-line distance  
to Bucharest

|                |     |
|----------------|-----|
| Arad           | 366 |
| Bucharest      | 0   |
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# A\* search example



$$f(n) = g(n) + h(n)$$

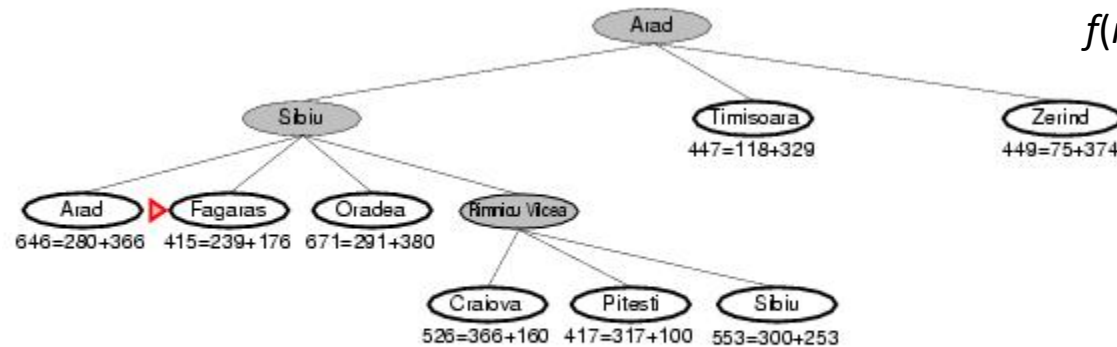


Straight-line distance to Bucharest

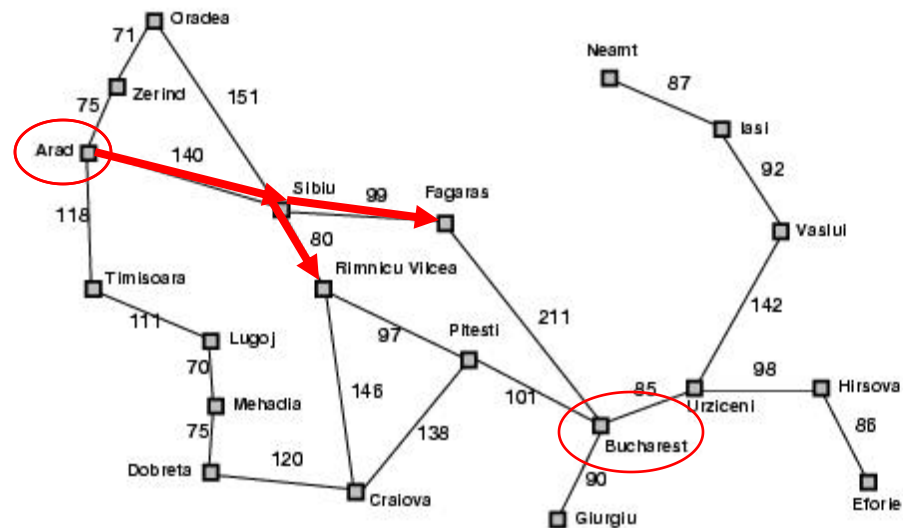
|                |     |
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# A\* search example



$$f(n) = g(n) + h(n)$$

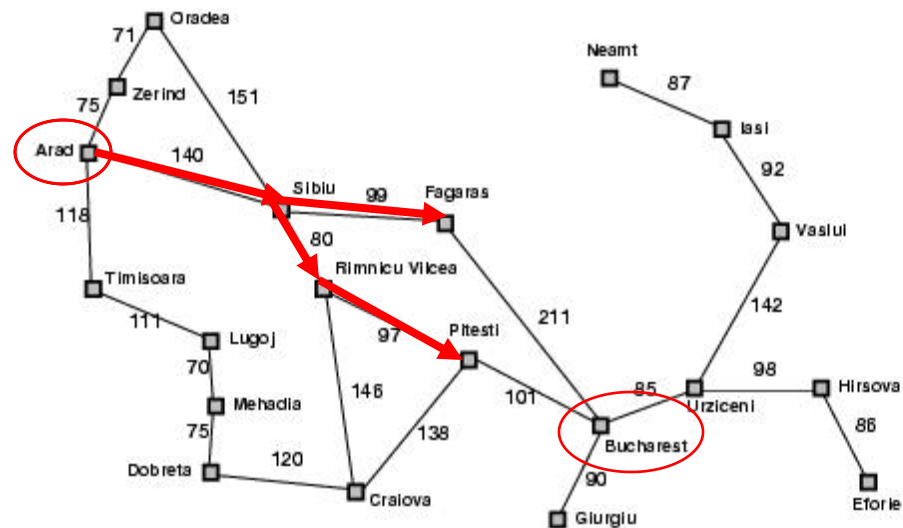
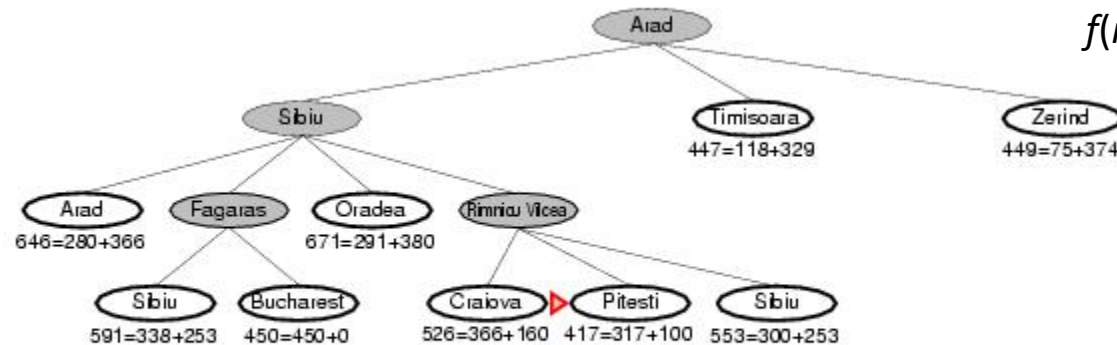


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# A\* search example

$$f(n) = g(n) + h(n)$$

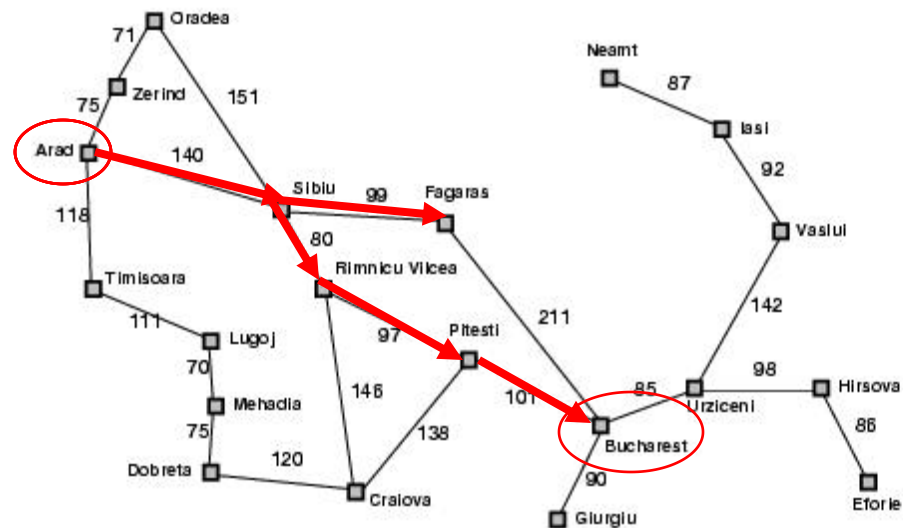
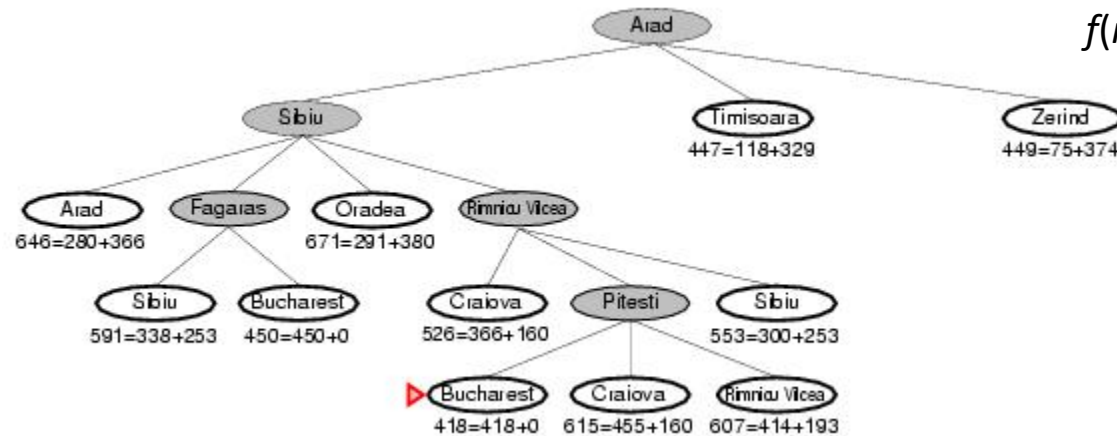


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# A\* search example

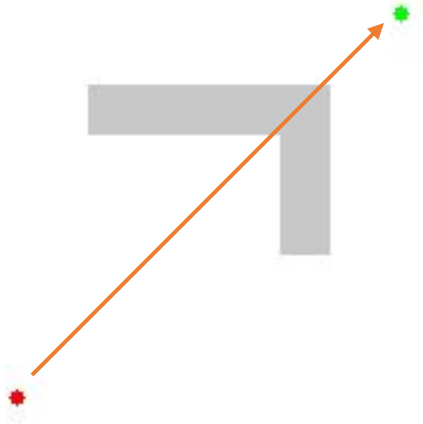
$$f(n) = g(n) + h(n)$$



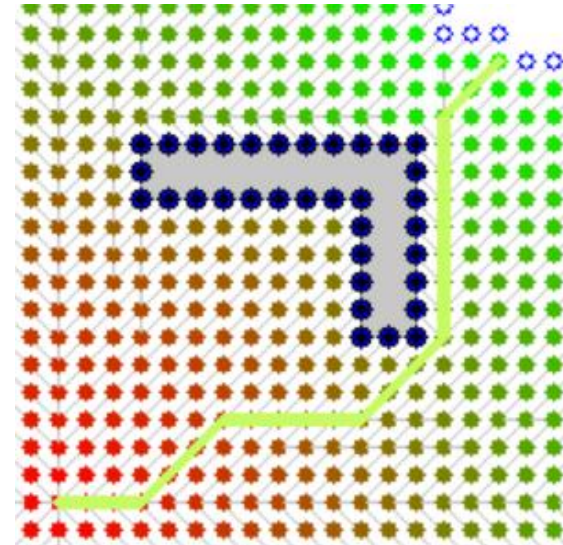
Straight-line distance to Bucharest

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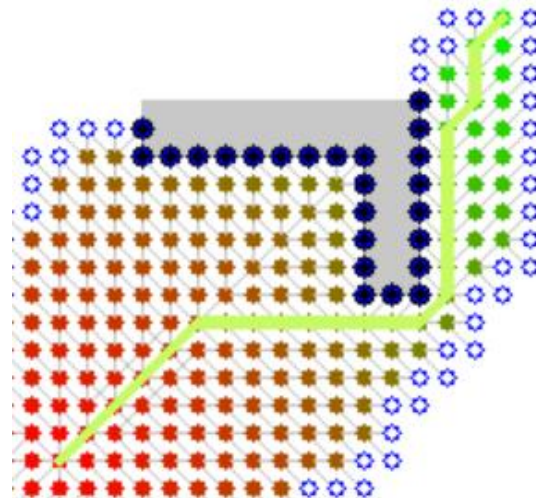
# BFS vs. A\* search



BFS



A\*



Source: [Wikipedia](https://en.wikipedia.org/wiki/Breadth-first_search)

# Implementation of A\* Search

Best-First  
Search



Expand the frontier  
using  
 $f(n) = g(n) + h(n)$

# Implementation of A\* Search

Path cost to  $n$  + heuristic from  $n$  to goal

$$g(n) + h(n)$$

```

function BEST-FIRST-SEARCH(problem, f) returns a solution node or failure
  node  $\leftarrow$  NODE(STATE=problem.INITIAL)
  frontier  $\leftarrow$  a priority queue ordered by f, with node as an element
  reached  $\leftarrow$  a lookup table, with one entry with key problem.INITIAL and value node
  while not IS-EMPTY(frontier) do
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      if s is not in reached or child.PATH-COST < reached[s].PATH-COST then
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        add child to frontier
  return failure
  
```

The order for expanding the frontier is determined by  $f(n)$

This check is the different to BFS! It visits a node again if it can be reached by a better (cheaper) path.

See BFS for function EXPAND.

# Optimality: Admissible heuristics

**Definition:** A heuristic  $h$  is **admissible** if for every node  $n$ ,  $h(n) \leq h^*(n)$ , where  $h^*(n)$  is the true cost to reach the goal state from  $n$ .

I.e., an admissible heuristic is a **lower bound** and never overestimates the true cost to reach the goal.

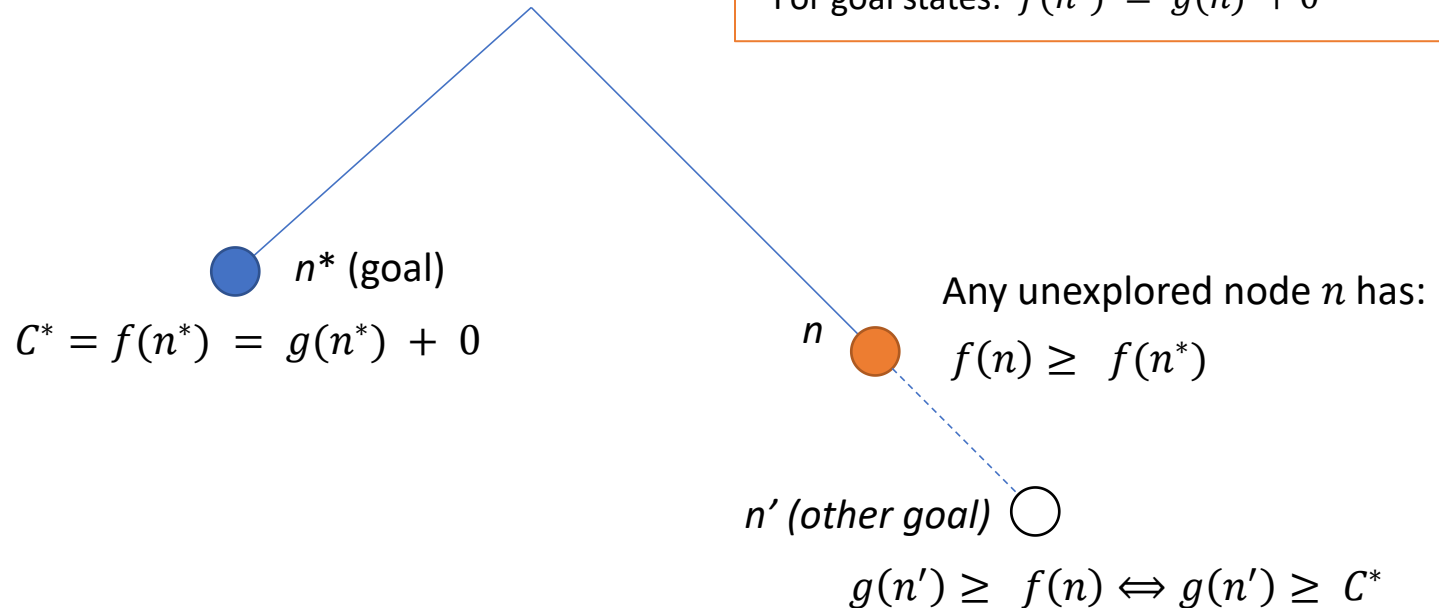
**Example:** straight line distance never overestimates the actual road distance.

**Theorem:** If  $h$  is admissible,  $A^*$  is optimal.

# Proof of Optimality of A\*

$$f(n) = g(n) + h(n)$$

For goal states:  $f(n^*) = g(n^*) + 0$



- Suppose A\* terminates its search at goal  $n^*$  at a cost of  $C^* = f(n^*)$ .
- All unexplored nodes  $n$  have  $f(n) \geq f(n^*)$  or they would have been explored before  $n^*$ .
- Since  $f(n)$  is an *optimistic* estimate, it is impossible for  $n$  to have a successor goal state  $n'$  with  $C' < C^*$ .
- This proves that  $n^*$  must be an optimal solution.



# Guarantees of $A^*$

$A^*$  is **optimally efficient**

- a. No other tree-based search algorithm that uses the same heuristic can expand fewer nodes and still be guaranteed to find the optimal solution.
- b. Any algorithm that does not expand all nodes with  $f(n) < C^*$  (the lowest cost of going to a goal node) cannot be optimal. It risks missing the optimal solution.

# Properties of $A^*$

- **Complete?**

Yes

- **Optimal?**

Yes

- **Time?**

Number of nodes for which  $f(n) \leq C^*$  (exponential)

- **Space?**

Same as time complexity.

# Designing heuristic functions

## Heuristics for the 8-puzzle

$h_1(n)$  = number of misplaced tiles

$h_2(n)$  = total Manhattan distance (number of squares from desired location of each tile)

|   |   |   |
|---|---|---|
| 7 | 2 | 4 |
| 5 |   | 6 |
| 8 | 3 | 1 |

Start State

|   |   |   |
|---|---|---|
|   | 1 | 2 |
| 3 | 4 | 5 |
| 6 | 7 | 8 |

Goal State

$$h_1(start) = 8$$

$$h_2(start) = 3 + 1 + 2 + 2 + 2 + 3 + 3 + 2 = 18$$

Are  $h_1$  and  $h_2$  admissible?

1 needs to move 3 positions

# Heuristics from relaxed problems

- A problem with fewer restrictions on the actions is called a relaxed problem.
- **The cost of an optimal solution to a relaxed problem is an admissible heuristic for the original problem.**
- If the rules of the 8-puzzle are relaxed so that a tile can move **anywhere**, then  $h_1(n)$  gives the shortest solution.
- If the rules are relaxed so that a tile can move to **any adjacent square**, then  $h_2(n)$  gives the shortest solution.

|   |   |   |
|---|---|---|
| 7 | 2 | 4 |
| 5 |   | 6 |
| 8 | 3 | 1 |

Start State

|   |   |   |
|---|---|---|
|   | 1 | 2 |
| 3 | 4 | 5 |
| 6 | 7 | 8 |

Goal State

$$h_1(start) = 8$$

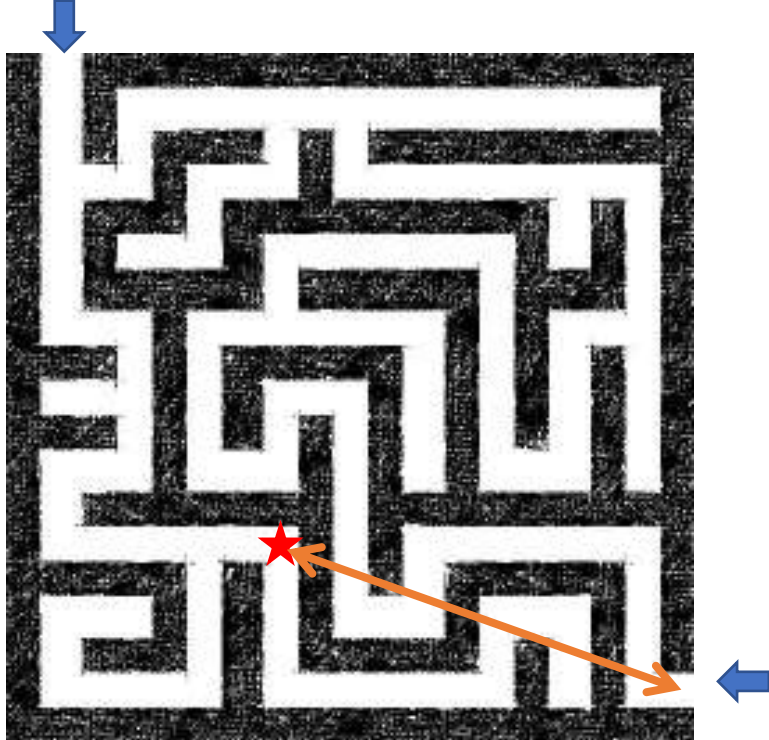
$$\begin{aligned}
 h_2(start) &= 3 + 1 + 2 + 2 + 2 + 3 + 3 + 2 \\
 &= 18
 \end{aligned}$$

# Heuristics from relaxed problems

What relaxations are used in these two cases?

**Euclidean distance**

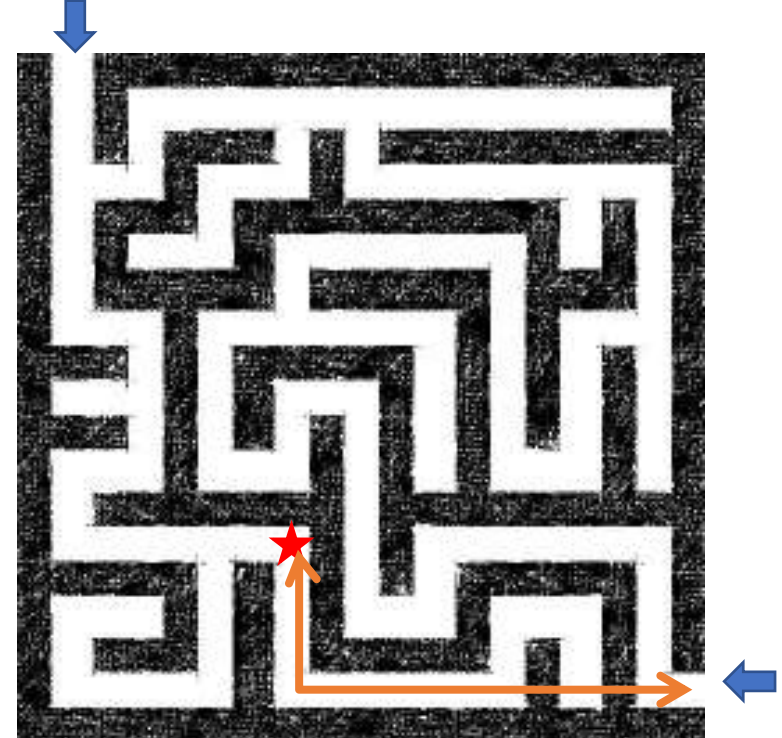
Start state



Goal state

**Manhattan distance**

Start state



Goal state

# Heuristics from subproblems

- Let  $h_3(n)$  be the cost of getting a subset of tiles (say, 1,2,3,4) into their correct positions. The final order of the \* tiles does not matter.
- Small subproblems are often easy to solve.
- Can precompute and save the exact solution cost for every or many possible subproblem instances – ***pattern database***.

|   |   |   |
|---|---|---|
| * | 2 | 4 |
| * |   | * |
| * | 3 | 1 |

Start State

|   |   |   |
|---|---|---|
|   | 1 | 2 |
| 3 | 4 | * |
| * | * | * |

Goal State

## Dominance: What heuristic is better?

**Definition:** If  $h_1$  and  $h_2$  are both admissible heuristics and  $h_2(n) \geq h_1(n)$  for all  $n$ , then  $h_2$  **dominates**  $h_1$

Which one is better for search?

- A\* search expands every node with  $f(n) < C^* \Leftrightarrow h(n) < C^* - g(n)$
- A\* search with  $h_2$  will expand less nodes and is therefore better.

# Dominance

- Typical search costs for the 8-puzzle (average number of nodes expanded for different solution depths  $d$ ):
  - $d = 12$     IDS    = 3,644,035 nodes  
                   $A^*(h_1)$  = 227 nodes  
                   $A^*(h_2)$  = 73 nodes
  - $d = 24$     IDS     $\approx$  54,000,000,000 nodes  
                   $A^*(h_1)$  = 39,135 nodes  
                   $A^*(h_2)$  = 1,641 nodes



# Combining heuristics

- Suppose we have a collection of admissible heuristics  $h_1, h_2, \dots, h_m$ , but none of them dominates the others.
- Combining them is easy:

$$h(n) = \max\{h_1(n), h_2(n), \dots, h_m(n)\}$$

- That is, always pick for each node the heuristic that is closest to the real cost to the goal  $h^*(n)$ .

# Satisficing Search: Weighted A\* search

- Often it is sufficient to find a **“good enough” solution** if it can be found very quickly or with way less computational resources.
- We could use inadmissible heuristics in A\* search (e.g., by multiplying  $h(n)$  with a factor  $W$ ) that sometimes overestimate the optimal cost to the goal slightly.
  1. It potentially reduces the number of expanded nodes significantly.
  2. This will break the algorithm’s optimality guaranty!

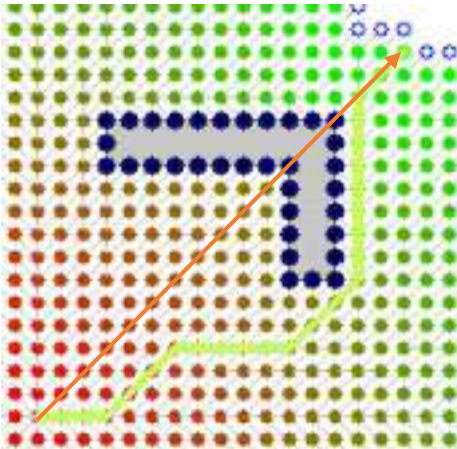
$$f(n) = g(n) + W \times h(n)$$

|                            |                        |                    |
|----------------------------|------------------------|--------------------|
| <b>Weighted A* search:</b> | $g(n) + W \times h(n)$ | $(1 < W < \infty)$ |
|----------------------------|------------------------|--------------------|

Other algorithms are special cases:

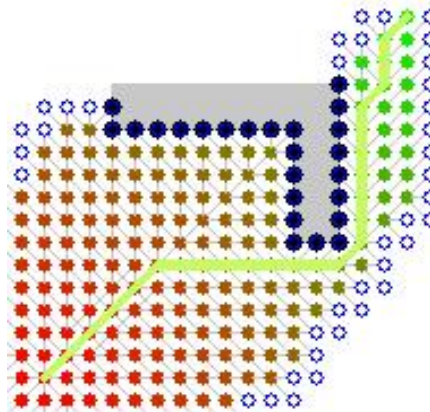
|                           |               |                |
|---------------------------|---------------|----------------|
| A* search:                | $g(n) + h(n)$ | $(W = 1)$      |
| Uniform cost search:      | $g(n)$        | $(W = 0)$      |
| Greedy best-first search: | $h(n)$        | $(W = \infty)$ |

# Example of weighted A\* search



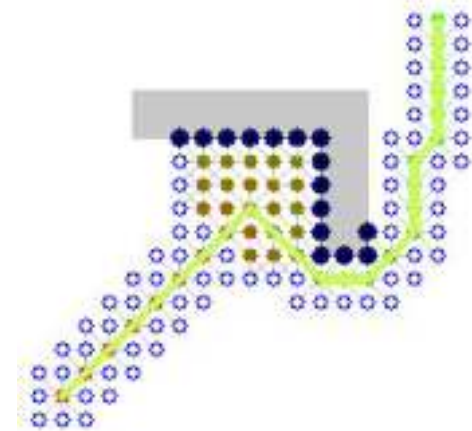
BFS

$$f(n) = \text{\# actions to reach } n$$



Exact A\* Search

$$f(n) = g(n) + h_{Eucl}(n)$$



Weighted A\* Search

$$f(n) = g(n) + 5 h_{Eucl}(n)$$

# Memory-bounded search

- The memory usage of  $A^*$  (search tree and frontier) can still be exorbitant.
- How can we make  $A^*$  more memory-efficient while maintaining completeness and optimality?
  - Iterative deepening  $A^*$  search.
  - Recursive best-first search, SMA\*: Forget some subtrees but remember the best  $f$ -value in these subtrees and regenerate them later if necessary.
- **Problems:** memory-bounded strategies can be complicated to implement and suffer from “memory thrashing” (regenerating forgotten nodes like IDS).

# Implementation as Best-first Search

- All discussed search strategies can be implemented using Best-first search.
- Best-first search expands always the **node with the minimum value of an evaluation function  $f(n)$** .

| Search Strategy          | Evaluation function $f(n)$  |
|--------------------------|-----------------------------|
| BFS                      | $g(n)$ (=uniform path cost) |
| Uniform-cost Search      | $g(n)$ (=path cost)         |
| DFS/IDS (see note below) | $-g(n)$                     |
| Greedy Best-first Search | $h(n)$                      |
| (weighted) A* Search     | $g(n) + W \times h(n)$      |

- **Note:** DFS/IDS is typically implemented differently to achieve the lower space complexity.

# Implementation: Best-First Search Strategy

This is the function  $f(n)$

```
function BEST-FIRST-SEARCH(problem, f) returns a solution node or failure
  node  $\leftarrow$  NODE(STATE=problem.INITIAL)
  frontier  $\leftarrow$  a priority queue ordered by f, with node as an element
  reached  $\leftarrow$  a lookup table, with one entry with key problem.INITIAL and value node
  while not IS-EMPTY(frontier) do
    node  $\leftarrow$  POP(frontier)
    if problem.IS-GOAL(node.STATE) then return node
    for each child in EXPAND(problem, node) do
      s  $\leftarrow$  child.STATE
      if s is not in reached or child.PATH-COST < reached[s].PATH-COST then
        reached[s]  $\leftarrow$  child
        add child to frontier
  return failure
```

The order for expanding the frontier is determined by  $f(n)$  = path cost to node  $n$ .

This check is the different to BFS! It visits a node again if it can be reached by a better (cheaper) path.

See BFS for function EXPAND.

# Summary: Uninformed search strategies

| Algorithm                  | Complete?                         | Optimal?                    | Time complexity                      | Space complexity |
|----------------------------|-----------------------------------|-----------------------------|--------------------------------------|------------------|
| <b>BFS</b>                 | Yes                               | If all step costs are equal | $O(b^d)$                             | $O(b^d)$         |
| <b>Uniform-cost Search</b> | Yes                               | Yes                         | Number of nodes with $g(n) \leq C^*$ |                  |
| <b>DFS</b>                 | In finite spaces (cycle checking) | No                          | $O(b^m)$                             | $O(bm)$          |
| <b>IDS</b>                 | Yes                               | If all step costs are equal | $O(b^d)$                             | $O(bd)$          |

b: maximum branching factor of the search tree  
 d: depth of the optimal solution  
 m: maximum length of any path in the state space  
 C\*: cost of optimal solution

# Summary: All search strategies

| Algorithm                       | Complete?                          | Optimal?                    | Time complexity                             | Space complexity                           |
|---------------------------------|------------------------------------|-----------------------------|---------------------------------------------|--------------------------------------------|
| <b>BFS</b>                      | Yes                                | If all step costs are equal | $O(b^d)$                                    | $O(b^d)$                                   |
| <b>Uniform-cost Search</b>      | Yes                                | Yes                         | Number of nodes with $g(n) \leq C^*$        |                                            |
| <b>DFS</b>                      | In finite spaces (cycles checking) | No                          | $O(b^m)$                                    | $O(bm)$                                    |
| <b>IDS</b>                      | Yes                                | If all step costs are equal | $O(b^d)$                                    | $O(bd)$                                    |
| <b>Greedy best-first Search</b> | In finite spaces (cycles checking) | No                          | Depends on heuristic                        | Worst case: $O(b^m)$<br>Best case: $O(bd)$ |
| <b>A* Search</b>                | Yes                                | Yes                         | Number of nodes with $g(n) + h(n) \leq C^*$ |                                            |



# Conclusion

- Tree search can be used for planning actions for **goal-based agents** in known, fully observable and deterministic environments.
- Issues are:
  - The large search space typically does not fit into memory. We use a description using a compact **transition model**.
  - The search tree is built on the fly, and we have to deal with **cycles and redundant paths**.
- IDS is a memory efficient methods used in AI often for **uninformed search**.
- **Informed search** uses heuristics based on knowledge or percepts to improve search.